

Panama — Ley 81 de 2019

Discovery and masking of Panamanian personal data: cédula and RUC with a valid check digit, quasi-identifiers, the sensitive data of article 4, economic and credit data, and criminal records.

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DOMAINS

135

CLASSIFIERS

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About this profile set

This profile set finds the personal data in a Panamanian database and masks it, so that a non-production copy — for testing, development, training or analytics — no longer identifies anyone.

Masking only names and cédulas does not get there: a birth date, a sex and a small corregimiento are enough to point at one person. The set works on the same layers the risk is in:

- **Direct identifiers** — cédula, RUC and its DV, names, contact, accounts and cards, passport and immigration documents, professional licenses, property, devices, vehicles, free text.
- **Quasi-identifiers** — birth date, age, sex, district, corregimiento, barriada, postal code, occupation, employer, schooling.
- **Sensitive data** — ethnic origin and indigenous language, health, biometric and genetic data, religious, philosophical and moral beliefs, political opinions and affiliation, union membership, sexual orientation.
- **Economic and credit data** — APC credit references, debt, salary, assets, pensions, social security coverage, social programs, poverty, housing tenure.
- **Criminal records** — police records and case numbers.

On a Masking Engine the set is one profile set with its classifiers, the domains they vote for, the algorithms those domains name — and the algorithms those build on — and the lookup files they read. The appendices list every one of them with its full configuration.

The assignment threshold is 60%. An unambiguous column name reaches it alone; an ambiguous one — NOMBRE , RESULTADO , CIUDAD — needs the values to agree. A column with a meaningless name (CAMP01 , OBS) is still found when its values have a recognizable shape or vocabulary: a cédula, a RUC with its DV, a plate, a corregimiento.

Column names are recognized in Spanish, in the abbreviations of Panamanian systems (NUM_CEDULA , CIP , FEC_NAC , COD_CORREG , NIS), in the fields of the electronic invoice (dRuc , dDV) and in English.

Before masking, inventory the binary columns (BLOBs, images, PDFs). Onboarding processes often store a photo of the cédula: the number is still readable in the image. Report those columns as residual risk.

Law 81 of 2019 at a glance

Law 81 of 26 March 2019, on Personal Data Protection, was published in Gaceta Oficial 28743-A of 29 March 2019 and has been in force since 29 March 2021, two years after its enactment (art. 47). Executive Decree 285 of 28 May 2021 regulates it. The supervisory authority is the **National Authority for Transparency and Access to Information (ANTAI)**, through its Personal Data Protection Directorate.

What matters for a copy of production data:

- **Personal data** is any information concerning natural persons that identifies them or makes them identifiable (art. 4, 9). The law protects natural persons: a legal entity's data is outside its scope.
- **Sensitive data** is data that refers to the data subject's intimate sphere, or whose misuse may give rise to discrimination or carry a serious risk. As examples, not as a closed list, the law names data that may reveal racial or ethnic origin; religious, philosophical and moral beliefs or convictions; union membership; political opinions; data concerning health, life, sexual preference or orientation, genetic data or biometric data (art. 4, 11). Decree 285 defines biometric, genetic and health data (art. 4, 2 to 4).

- **Anonymized data is out of scope.** Processing of information obtained through a prior dissociation or anonymization procedure, such that the result cannot be associated with the data subject, is excluded from the law (art. 3, 5). *Dissociated data* cannot be associated with the data subject nor, by its structure, content or level of detail, allow their identification (art. 4, 10).
- **Purpose.** Data is used for the purposes it was authorized for; any other use needs consent, a special law or another legal basis (arts. 2, 2 and 11). Further processing for research or studies needs no consent when the data is anonymized (art. 12).
- **Health and sensitive data.** Consent for health data must be prior, irrefutable and express (art. 8; Decree 285, art. 18). Sensitive data may not be transferred save for exceptions, and for historical, statistical or scientific purposes it must be dissociated (art. 13).
- **Economic, financial, banking or commercial data** is processed with prior consent (art. 8, 3). Credit references also have a law of their own, Law 24 of 2002.
- **Criminal convictions** have their own disclosure rules (art. 30; Decree 285, art. 7).
- **Security.** Proportionate technical and organizational measures, especially for sensitive data (art. 2, 5; Decree 285, art. 36). A security breach is notified to ANTAI and to the affected data subjects within seventy-two hours (Decree 285, art. 37).
- **Sanctions.** Fines from one thousand to ten thousand balboas set by ANTAI (art. 36). Disregarding the rules on sensitive data is a very serious infringement, which can lead to closing the database's records or suspending the processing activity (arts. 41 and 43).

A test environment with production data is processing for a purpose the data subject did not authorize, and it holds sensitive and financial data outside the controls built for them. A copy that is effectively anonymized falls, under article 3, outside the law: that is what the set is for.

The list of sensitive data is not exhaustive. The set also treats gender identity and sexual life as sensitive, since they expose people to discrimination in the same way. Reform proposals are under discussion, among them higher fines; this document follows the text in force in September 2026.

This document describes technical controls. Whether a copy counts as anonymized is a legal assessment of the whole environment — access, retention, what else it is combined with — not of the profile set alone.

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How sensitive data is discovered

A profiling job looks at every column of its rule set and asks each classifier of the profile set how confident it is that the column belongs to its domain. The set uses the four classifier frameworks of the Masking Engine:

- **Column name** (Path) — regular expressions on the column name. Each rule has a match strength: the confidence it gives when it matches.
- **Data type** (Type) — accepts the column type and length, or rules the domain out. It never assigns a domain on its own.
- **Value format** (Regex) — tests a sample of the values against patterns, some with a check digit. The confidence follows the share of values that match.
- **Known values** (List) — looks the sampled values up in lists; for names, word by word.

Evidence from the name and from the values is combined per domain, and the column is assigned to the domain with the highest confidence when it reaches the assignment threshold of the profile set — **60%** here. Values that do not match a format or a list count against their domain by the classifier's reject strength.

Column-name rules are shown as their alternatives. They ignore case and must not be glued to other letters:

`rut` finds `RUT_CLIENTE` and `CLIENTE_RUT` but not `RUTA`. In them, `_?` is an optional underscore, `(a|b)`

either form, `s?` an optional letter and `[a-z]*` any continuation.

Profiling proposes; it does not decide. Review the domains assigned in the inventory before running a masking job, and look for columns the set left unassigned.

4

How data is masked

Each domain names one algorithm, which a masking job applies to every column the inventory assigns to the domain. Algorithms can be composed: a **Regex Decompose** splits a value into parts and gives each part its own action; a **String Algorithm Chain** applies algorithms one after another. The tree under each domain shows everything its algorithm uses; Appendix A has every configuration in full.

The algorithms are deterministic for their key: the same value masks the same way in every table, database and job, so joins between masked systems keep working — as long as every engine uses the same keys.

The masked examples in this document were produced with one particular key. On an engine with other keys the masked values differ; their shape does not.

5

Essential and extended packs

The set can be adopted whole or in part. The **extended pack** is everything this document describes. The **essential pack** is the minimum for the law — identity documents, names, contact details, address and birth date. To adopt only the essential pack, create on the engine just its domains, the classifiers that vote for them and the algorithms and files those use, and put only those classifiers in the profile set.

PACK	DOMAINS	CLASSIFIERS	ALGORITHMS	FILES
Essential	12	27	59	10
Extended	76	135	142	79

The essential pack holds these domains, marked **Essential** in the tables and chapters:

- Cédula (national ID) — `PA_L1_CEDULA`
- RUC (taxpayer number) — `PA_L1_RUC`
- DV in its own column — `PA_L1_RUC_DV`
- Passport — `PA_L1_PASAPORTE`
- Given names — `PA_L1_NOMBRE`
- Surnames — `PA_L1_APELLIDO`
- Full name — `PA_L1_NOMBRE_COMPLETO`
- E-mail — `PA_L1_EMAIL`
- Phone — `PA_L1_TELEFONO`
- Address — `PA_L1_DIRECCION`
- Address detail — `PA_L1_DIRECCION_COMPLEMENTO`
- Birth date — `PA_L2_FECHA_NACIMIENTO`

With fewer domains competing, a column that a domain of the extended pack would claim by its name can be proposed for an essential domain instead — a numeric code, for instance, can look like a telephone number. Review the profiling result.

Domains at a glance

Every domain of the set, how it is found and the algorithm that masks it.

DOMAIN	FOUND BY			ALGORITHM	
DIRECT IDENTIFIERS					
Cédula (national ID)	ESSENTIAL	COLUMN NAME	DATA TYPE	VALUE FORMAT	PA_CEDULA
PA_L1_CEDULA					
RUC (taxpayer number)	ESSENTIAL	COLUMN NAME	DATA TYPE	VALUE FORMAT	PA_RUC
PA_L1_RUC					
DV in its own column	ESSENTIAL	COLUMN NAME	DATA TYPE		PA_DV
PA_L1_RUC_DV					
Passport	ESSENTIAL	COLUMN NAME	VALUE FORMAT		PA_CM_ALFANUM
PA_L1_PASAPORTE					
Immigration document		COLUMN NAME			PA_CM_ALFANUM
PA_L1_DOCUMENTO_MIGRATORIO					
Social security number		COLUMN NAME			PA_CM_ALFANUM
PA_L1_SEGURO_SOCIAL					
Professional license (idoneidad)		COLUMN NAME			PA_CM_ALFANUM
PA_L1_IDONEIDAD					
Given names	ESSENTIAL	COLUMN NAME	KNOWN VALUES		PA_NOMBRES
PA_L1_NOMBRE					
Surnames	ESSENTIAL	COLUMN NAME	KNOWN VALUES		PA_APELLIDOS
PA_L1_APELLIDO					
Full name	ESSENTIAL	COLUMN NAME	KNOWN VALUES		PA_NOMBRE_COMPLETO
PA_L1_NOMBRE_COMPLETO					
E-mail	ESSENTIAL	COLUMN NAME	DATA TYPE	VALUE FORMAT	PA_EMAIL
PA_L1_EMAIL					
Phone	ESSENTIAL	COLUMN NAME	VALUE FORMAT		PA_TELEFONO
PA_L1_TELEFONO					
Address	ESSENTIAL	COLUMN NAME	DATA TYPE	VALUE FORMAT	PA_DIRECCION
PA_L1_DIRECCION					
Address detail	ESSENTIAL	COLUMN NAME			PA_CM_ALFANUM
PA_L1_DIRECCION_COMPLEMENTO					
Bank account		COLUMN NAME			PA_CM_ALFANUM
PA_L1_CUENTA_BANCARIA					
Payment card		COLUMN NAME	VALUE FORMAT		PA_TARJETA
PA_L1_TARJETA					
IP address		COLUMN NAME	VALUE FORMAT		PA_IP
PA_L1_IP					
Device identifier		COLUMN NAME	VALUE FORMAT		PA_CM_HEX
PA_L1_DISPOSITIVO					
License plate		COLUMN NAME	VALUE FORMAT		PA_PLACA
PA_L1_PLACA					

DOMAIN	FOUND BY			ALGORITHM
Vehicle identification number PA_L1_VIN	COLUMN NAME	VALUE FORMAT		PA_CM_ALFANUM
Property number (finca) PA_L1_FINCA	COLUMN NAME			PA_CM_ALFANUM
Contract or customer number PA_L1_NUMERO_CONTRATO	COLUMN NAME			PA_CM_ALFANUM
User name or social network PA_L1_USUARIO_RRSS	COLUMN NAME			PA_CM_ALFANUM
Credential PA_L1_CREDENCIAL	COLUMN NAME			PA_REDACTAR
Geolocation PA_L1_GEOLOCALIZACION	COLUMN NAME	VALUE FORMAT		PA_COORDENADA
Free text PA_L1_TEXTO_LIBRE	COLUMN NAME	DATA TYPE	VALUE FORMAT	PA_TEXTO_LIBRE
QUASI-IDENTIFIERS				
Birth date ESSENTIAL PA_L2_FECHA_NACIMIENTO	COLUMN NAME	DATA TYPE		PA_FECHA_NACIMIENTO
Birth year PA_L2_ANIO_NACIMIENTO	COLUMN NAME	DATA TYPE		PA_ANIO
Age PA_L2_EDAD	COLUMN NAME	DATA TYPE		PA_EDAD
Life event dates PA_L2_FECHA_EVENTO	COLUMN NAME	DATA TYPE		PA_FECHA_EVENTO
Sex PA_L2_SEXO	COLUMN NAME	KNOWN VALUES		PA_SEXO
District PA_L2_DISTRITO	COLUMN NAME	KNOWN VALUES		PA_DISTRITO
Corregimiento PA_L2_CORREGIMIENTO	COLUMN NAME	KNOWN VALUES		PA_CORREGIMIENTO
Corregimiento or district code PA_L2_CODIGO_LUGAR	COLUMN NAME	DATA TYPE	KNOWN VALUES	PA_CODIGO_LUGAR
Populated place PA_L2_LUGAR_POBLADO	COLUMN NAME			PA_LUGAR_POBLADO
Barriada or neighborhood PA_L2_BARRIADA	COLUMN NAME			PA_BARRIADA
Postal code PA_L2_CODIGO_POSTAL	COLUMN NAME	VALUE FORMAT		PA_CODIGO_POSTAL
Nationality PA_L2_NACIONALIDAD	COLUMN NAME	KNOWN VALUES		PA_NACIONALIDAD
Marital status PA_L2_ESTADO_CIVIL	COLUMN NAME	KNOWN VALUES		PA_ESTADO_CIVIL
Occupation PA_L2_OCUPACION	COLUMN NAME			PA_OCUPACION
Employer PA_L2_EMPLEADOR	COLUMN NAME			PA_EMPLEADOR

DOMAIN	FOUND BY		ALGORITHM
Schooling PA_L2_ESCOLARIDAD	COLUMN NAME	KNOWN VALUES	PA_ESCOLARIDAD
School PA_L2_ESCUELA	COLUMN NAME		PA_ESCUELA
Children and dependents PA_L2_DEPENDIENTES	COLUMN NAME	DATA TYPE	PA_CANTIDAD_ACOTADA
SENSITIVE DATA			
Ethnic origin PA_L3_ETNIA	COLUMN NAME	KNOWN VALUES	PA_PUEBLO
Indigenous language PA_L3 LENGUA_INDIGENA	COLUMN NAME	KNOWN VALUES	PA LENGUA_INDIGENA
Diagnosis PA_L3_SALUD_DIAGNOSTICO	COLUMN NAME	VALUE FORMAT	PA_DIAGNOSTICO
Medication PA_L3_SALUD_MEDICAMENTO	COLUMN NAME	KNOWN VALUES	PA_MEDICAMENTO
Clinical identifier PA_L3_SALUD_IDENTIFICADOR	COLUMN NAME		PA_CM_ALFANUM
Clinical text PA_L3_SALUD_TEXTO_CLINICO	COLUMN NAME	DATA TYPE	PA_TEXTO_LIBRE
Test result PA_L3_SALUD_RESULTADO_EXAMEN	COLUMN NAME		PA_CM_ALFANUM
Blood group PA_L3_SALUD_GRUPO_SANGUINEO	COLUMN NAME	KNOWN VALUES	PA_GRUPO_SANGUINEO
Disability PA_L3_SALUD_DISCAPACIDAD	COLUMN NAME	KNOWN VALUES	PA_DISCAPACIDAD
Sexual and reproductive health PA_L3_SALUD_SEXUAL_REPRODUCTIVA	COLUMN NAME		PA_CATEGORIA_SUPRIMIDA
Mental health and addiction PA_L3_SALUD_MENTAL_ADICCIONES	COLUMN NAME		PA_CATEGORIA_SUPRIMIDA
Biometric data PA_L3_BIOMETRICO	COLUMN NAME	DATA TYPE	PA_REDACTAR
Genetic data PA_L3_INFORMACION_GENETICA	COLUMN NAME		PA_REDACTAR
Sexual orientation PA_L3_ORIENTACION_SEXUAL	COLUMN NAME	KNOWN VALUES	PA_ORIENTACION_SEXUAL
Gender identity PA_L3_IDENTIDAD_GENERO	COLUMN NAME	KNOWN VALUES	PA_IDENTIDAD_GENERO
Sexual life PA_L3_VIDA_SEXUAL	COLUMN NAME		PA_CATEGORIA_SUPRIMIDA
Religion PA_L3_RELIGION	COLUMN NAME	KNOWN VALUES	PA_RELIGION
Philosophical and moral beliefs PA_L3_CREENCIA_FILOSOFICA_MORAL	COLUMN NAME		PA_CATEGORIA_SUPRIMIDA
Political opinion PA_L3_OPINION_POLITICA	COLUMN NAME	KNOWN VALUES	PA_OPINION_POLITICA

DOMAIN	FOUND BY		ALGORITHM
Political affiliation PA_L3_AFILIACION_POLITICA	COLUMN NAME	KNOWN VALUES	PA_PARTIDO_POLITICO
Union membership PA_L3_AFILIACION_SINDICAL	COLUMN NAME	KNOWN VALUES	PA_CATEGORIA_SUPRIMIDA
ECONOMIC AND CREDIT DATA			
Credit reference PA_FIN_REFERENCIA_CREDITO	COLUMN NAME		PA_APC
Debt amount PA_FIN_MONTO_DEUDA	COLUMN NAME	DATA TYPE	PA_MONTO_DEUDA
Salary and income PA_FIN_INGRESO_MENSUAL	COLUMN NAME	DATA TYPE	PA_INGRESO_MENSUAL
Assets PA_FIN_PATRIMONIO	COLUMN NAME	DATA TYPE	PA_PATRIMONIO
Pension and individual account PA_FIN_PENSION	COLUMN NAME	DATA TYPE	PA_MONTO_PENSION
Social security coverage PA_FIN_SEGURO_SOCIAL	COLUMN NAME	KNOWN VALUES	PA_SEGURO_SOCIAL
Social program PA_FIN_PROGRAMA_SOCIAL	COLUMN NAME	KNOWN VALUES	PA_PROGRAMA_SOCIAL
Socioeconomic level PA_FIN_NIVEL_SOCIOECONOMICO	COLUMN NAME	KNOWN VALUES	PA_NIVEL_SOCIOECONOMICO
Housing tenure PA_FIN_VIVIENDA	COLUMN NAME	KNOWN VALUES	PA_VIVIENDA
CRIMINAL RECORDS			
Criminal record PA_PENAL_ANTECEDENTES	COLUMN NAME		PA_ANTECEDENTES
Case number PA_PENAL_CAUSA	COLUMN NAME	VALUE FORMAT	PA_CM_ALFANUM

Direct identifiers

Data that identifies a person on its own. The cédula is the key of almost every Panamanian record: it is a natural person's RUC, the number of their driver's license and their identification at the Social Security Fund. It is masked the same way everywhere, and the RUC keeps a valid check digit, because applications validate it.

7.1 Cédula (national ID)

PA_L1_CEDULA

- ESSENTIAL
- COLUMN NAME
- DATA TYPE
- VALUE FORMAT
- REGEX DECOMPOSE

WHAT IT IS

The number of the identity card issued by the Electoral Tribunal: province of birth (1 to 13; 10, 11 and 12 are the Guna Yala, Emberá-Wounaan and Ngäbe-Buglé comarcas), tomo and asiento of the civil registry entry. 8–442–445 . Letters mark other cases: E resident foreigner, N naturalized, PE Panamanian born abroad, AV registered before the civil registry law, PI indigenous Panamanian; 8–NT–12–3456 is the tax number of someone without a cédula. In Panama the driver's license carries the cédula number, so license columns fall here too.

HOW IT IS FOUND

COLUMN NAME PA_L1_CEDULA – Path

- 90%cedula[a-z0-9]* c[eé]dula ced cip n(o|um|ro|umero)?(de_)?cedula cedula_(identidad|personal|cliente|empleado|paciente|asegurado|beneficiario|titular|conyuge|acudiente)(cliente|empleado|paciente|asegurado|beneficiario|titular)?cedula documento?(de_)?identidad doc?identidad id?personal identificacion?personal n(o|um|ro)?identificacion numero?documento num?doc nro?doc national?id dni
- 80%licencia?(de_)?(conducir|manejo|chofer|conductor) n(o|um|ro)?licencia(_?conducir)?licencia driver?license drivers?license

DATA TYPE PA_L1_CEDULA – Type

Accepts text (length 5 or more); any other type rules the domain out.

VALUE FORMAT PA_L1_CEDULA – Regex

PATTERN	STRENGTH	NOTES
(?!\\d{1,2}-\\d{1,2}-(19 20)\\d{2}\$)(([1-9] 1[0-3])(AV PI)? E N PE)-\\d{1,4}-\\d{1,6}	100%	
([1-9] 1[0-3])-NT-\\d{1,4}-\\d{1,6}	90%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

The province becomes another province between 1 and 13, and tomo and asiento are mapped digit by digit keeping their length; the letters stay, and so does the document type. The mapping is one-to-one: a cédula used as a key does not collide. The same algorithm masks the cédula inside the RUC, so a person's cédula and RUC still agree. Text that is not shaped like a cédula is masked character by character.

PA_CEDULA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

- PA_PROVINCIA_CEDULA NUMERIC MAPPING
Maps the number to another between 1 and 13.
- PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
8-442-445	→	1-433-300
E-8-123456	→	E-1-904292
4PI-234-123	→	8PI-158-957

7.2 RUC (taxpayer number)

PA_L1_RUC

ESSENTIAL

COLUMN NAME

DATA TYPE

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

The DGI's Single Taxpayer Registry number. A natural person's RUC is their cédula, with a two-digit check digit next to it: 8-442-445 DV 08 . A legal entity's RUC is its Public Registry number: 155596713-2-2015 DV 59 . One column often mixes both.

HOW IT IS FOUND

COLUMN NAME PA_L1_RUC - Path

90%

```
ruc[a-z0-9]* d_?ruc n(o|um|ro|umero)?(de_)?ruc registro_?unico(_?de)?(_?
contribuyentes?)? ruc_?
(cliente|proveedor|emisor|receptor|contribuyente|persona|natural|empleado|beneficiario|titular
) (cliente|proveedor|emisor|receptor)?ruc tax_id id_fiscal nit
```

DATA TYPE PA_L1_RUC - Type

Accepts text (length 5 or more); any other type rules the domain out.

VALUE FORMAT PA_L1_RUC - Regex

PATTERN

STRENGTH NOTES

(?:([1-9]|1[0-3])(?:AV|PI)?-\d{1,4}-\d{1,6}|(?:E|N|PE)-\d{1,4}-\d{1,6})(?: +DV
+| +DV|-| +)\d{2}

100%

(?!\\d{1,2}-\\d{1,2}-(19|20)\\d{2}\$)(([1-9]|1[0-3])(AV|PI)?|E|N|PE)-\\d{1,4}-\\d{1,6}

60%

\\d{3,10}-\\d{1,4}-\\d{1,7}(\\s*DV\\s*\\d{2})?

60%

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

A natural person's RUC is masked as their cédula and the DV is recomputed, so every masked RUC passes DGI validation. The separators DV , DV , - and a space are recognized. A legal entity's RUC stays as it is: it is not personal data under article 4, and keeping it keeps joins with supplier catalogs working. See *The RUC check digit, computed by the set*.

PA_RUC REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_RUC_DV STRING ALGORITHM CHAIN

Applies PA_RUC_DV_1 → PA_RUC_DV_2 → PA_RUC_DV_3 → PA_RUC_DV_4 → PA_RUC_DV_5 , in this order. It is made of 46 algorithms that repeat the same step position by position; each one is in Appendix A.

PA_CEDULA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

EXAMPLES

VALUE		MASKED
8-442-445 DV 08	→	1-433-300 DV 12
1AV-432-658 DV 96	→	9AV-920-061 DV 24
155596713-2-2015 DV 59	→	155596713-2-2015 DV 59

7.3 DV in its own column

PA_L1_RUC_DV

ESSENTIAL

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

The RUC check digit kept apart, as in the electronic invoice (dDV) or a DV column.

HOW IT IS FOUND

COLUMN NAME

PA_L1_RUC_DV – Path

90%

dv d_?dv digito_?verificador dig_?verif dv_?ruc ruc_?dv check_?digit

DATA TYPE

PA_L1_RUC_DV – Type

Accepts text (length up to 3) or number (length up to 2); any other type rules the domain out.

HOW IT IS MASKED

The two digits are mapped. They cannot be recomputed: an algorithm sees one column, and the RUC is in another. Masking it keeps it from helping link the row to the real RUC.

PA_DV REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
08	→	57

7.4 Passport

PA_L1_PASAPORTE

ESSENTIAL

COLUMN NAME

VALUE FORMAT

CHARACTER MAPPING

WHAT IT IS

Panamanian and foreign passport numbers.

HOW IT IS FOUND

COLUMN NAME

PA_L1_PASAPORTE – Path

85%

pasaporte[a-z_]* passport[a-z_]* n(o|um|ro|umero)?pasaporte pasap

VALUE FORMAT

PA_L1_PASAPORTE – Regex

PATTERN	STRENGTH	NOTES
[A-Z]{1,2}\d{6,8}		30%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

PA0123456

→

MASKED

PU0730277

7.5 Immigration document

PA_L1_DOCUMENTO_MIGRATORIO

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Resident cards, residence and work permits, visas and foreign identification numbers. A permanent resident's E cédula is in the cédula domain.

HOW IT IS FOUND

COLUMN NAME

PA_L1_DOCUMENTO_MIGRATORIO - Path

80%

carne_?(migratorio|de_?migracion|residente|de_?residente|residencia|refugiado|trabajo)
carnet_?(migratorio|residente|residencia) permiso_?(de_?)(residencia|trabajo|migratorio)
n(o|u|m|ro)_?carne(t)? documento_?migratorio doc_?migratorio filiacion_?migratoria visa
n(o|u|m|ro)_?visa id_?extranjero documento_?extranjero identificacion_?extranjera

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

123456789

→

MASKED

110858568

7.6 Social security number

PA_L1_SEGURO_SOCIAL

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

The insured person's or card number of the Social Security Fund (CSS) kept by older systems. Today the CSS identifies the insured by their cédula.

HOW IT IS FOUND

COLUMN NAME

PA_L1_SEGURO_SOCIAL - Path

85%

n(o|u|m|ro|umero)_?(de_?)?seguro_?social seguro_?social_?(num|no|nro|numero) nss
n(o|u|m|ro)_?css css_?(num|no|numero|asegurado) n(o|u|m|ro)_?asegurado num_?asegurado
n(o|u|m|ro)_?(de_?)?afiliado(_?css)? carne_?(del_?)?seguro(_?social)? ss_?number social_?
security

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
123-4567	→	564-9285

7.7 Professional license (idoneidad)

PA_L1_IDONEIDAD

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

The idoneidad number that allows someone to practice a regulated profession — medicine, nursing, law, engineering. The registries are public: the number leads straight to the person's name.

HOW IT IS FOUND

COLUMN NAME PA_L1_IDONEIDAD - Path

85% idoneidad n(o|um|ro|umero)?(de_)?idoneidad registro?(medico|profesional|de_?idoneidad) licencia_?profesional codigo?(medico|profesional) colegiatura n(o|um|ro)?registro_? profesional

HOW IT IS MASKED

Character by character, keeping the length.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
12345	→	31564

7.8 Given names

PA_L1_NOMBRE

ESSENTIAL COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

First and middle names (NOMBRES , PRIMER_NOMBRE), including compound names such as Luis Carlos .

HOW IT IS FOUND

COLUMN NAME PA_L1_NOMBRE - Path

85% primer_?nombre segundo_?nombre nombres?_(de_)?pila nombre_?usual first_?name given_? names? fname name_?first middle_?name nombre[12] nombres (except when followed by list E1) nom_? (pila|trab|emp)

50% nombre (except when followed by list E2) nomb? (except when followed by list E3) name (except when followed by list E4)

KNOWN VALUES PA_L1_NOMBRE – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-nombres.txt	291	70%	María, Maria, José, Jose, Luis, Carlos

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each word is replaced by a name from a list of 253 names used in Panama, keeping the number of words and the case. More than three words are masked character by character.

PA_NOMBRES **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_NOMBRE_PILA **NAME**

Replaces the name with one of the 253 names in pa-nombres.txt .

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
Luis Carlos	→	Teodoro Norberto
YAMILETH	→	ZULEIKA

7.9 Surnames

PA_L1_APELLIDO

ESSENTIAL **COLUMN NAME** **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

Paternal and maternal surnames, in one column or two, and a married woman's surname: González de Pérez .

HOW IT IS FOUND

COLUMN NAME PA_L1_APELLIDO – Path

85%
apellidos? apellido?(paterno|materno|pat|mat|uno|dos|casada|de_?casada|[12]) ape_?
(pat|mat|paterno|materno|casada|[12]) apepat apemat ap_?(pat|mat|paterno|materno) a_
paterno a_?materno paterno materno primer_?apellido segundo_?apellido apellido_
usual last_?name surname lname family_?name second_?last_?name married_?name

KNOWN VALUES PA_L1_APELLIDO – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-apellidos.txt	258	70%	Rodríguez, Rodriguez, González, Gonzalez, Sánchez, Sanchez

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each surname is replaced by one from a list of 200 surnames common in Panama — led by Rodríguez, González, Sánchez, Martínez, Castillo and Ábrego —, keeping the number of words and the case. The de of a married surname stays.

PA_APELLIDOS **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_APELLIDO NAME

Replaces the name with one of the 200 names in `pa-apellidos.txt`.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

González de Pérez

→ Tapia de Aguilar

7.10 Full name

PA_L1_NOMBRE_COMPLETO

ESSENTIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Given names and surnames in one column, in Panamanian order, with `de` and the husband's surname where present, or as `APELLIDOS`, `NOMBRES`. Also the name of an account holder, beneficiary, insured person or guardian (*acudiente*).

HOW IT IS FOUND

COLUMN NAME PA_L1_NOMBRE_COMPLETO - Path

90%	<code>nombre_?(completo compl full)</code> <code>nombres?_?y_?apellidos?</code> <code>nombres?_?apellidos?</code> <code>nombreaapellido</code> <code>nom_?completo</code> <code>nomcompleto</code> <code>nombrecompleto</code> <code>full_?name</code> <code>fullname</code> <code>person_?</code> <code>name</code> <code>customer_?name</code> <code>employee_?name</code> <code>patient_?name</code> <code>nombre_?</code> <code>(cliente cte cli paciente pac trabajador trab empleado colaborador funcionario servidor_?</code> <code>publico alumno estudiante afiliado asegurado cotizante jubilado pensionado beneficiario benef</code> <code>titular madre padre acudiente tutor conyuge pareja medico doctor profesional responsable conta</code> <code>cto emergencia fiador codeudor deudor remitente destinatario receptor firmante testigo represe</code> <code>ntante apoderado propietario arrendatario inquilino contratante victima denunciante imputado s</code> <code>ocio donante heredero persona pers conductor chofer vendedor ejecutivo agente solicitante ciud</code> <code>adano elector votante contribuyente jefe_?(de_?)?hogar hijo hija familiar dependiente)</code>
60%	<code>titular</code> <code>beneficiario</code> <code>acudiente</code> <code>contacto_?emergencia</code> <code>conyuge</code> <code>tutor_?legal</code> <code>representante_?legal</code> <code>apoderado</code> <code>paciente</code> <code>victima</code> <code>asegurado</code> <code>contratante</code> <code>fiador</code> <code>codeudor</code> <code>jubilado</code> <code>pensionado</code>
50%	<code>nombre</code> (except when followed by <i>list E2</i>) <code>name</code> (except when followed by <i>list E4</i>)

KNOWN VALUES PA_L1_NOMBRE_COMPLETO - List

LIST	VALUES	STRENGTH	FIRST VALUES
<code>pa-detectar-nombres.txt</code>	291	60%	María, Maria, José, Jose, Luis, Carlos
<code>pa-detectar-apellidos.txt</code>	258	60%	Rodríguez, Rodriguez, González, Gonzalez, Sánchez, Sanchez

Values are split into words, and each word is looked up.

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Given names become given names and surnames become surnames, by position: one word is a given name, two are a given name and a surname, three are a given name and two surnames, four are two given names

and two surnames. `María Elena González de Pérez` keeps its `de` and its structure. A comma separates surnames from given names.

`PA_NOMBRE_COMPLETO` **REGEX DECOMPOSE**

Tries 8 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_APELLIDOS` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_APELLIDO` **NAME**

Replaces the name with one of the 200 names in `pa-apellidos.txt` .

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_NOMBRES` **REGEX DECOMPOSE**

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_NOMBRE_PILA` **NAME**

Replaces the name with one of the 253 names in `pa-nombres.txt` .

EXAMPLES

VALUE		MASKED
María Elena González de Pérez	→	Édgar Evelyn Tapia de Aguilar
SÁNCHEZ ÁBREGO, LUIS CARLOS	→	VALDÉS BATISTA, TEODORO NORBERTO

7.11 E-mail

`PA_L1_EMAIL`

`ESSENTIAL` `COLUMN NAME` `DATA TYPE` `VALUE FORMAT` `REGEX DECOMPOSE`

WHAT IT IS

Personal and work e-mail addresses.

HOW IT IS FOUND

`COLUMN NAME` `PA_L1_EMAIL` - Path

85%

`e_?mail[a-z0-9_]*` `mail[a-z0-9_]*` `correo(_?electronico)?[a-z0-9_]*` `email_?address`
`correo_?(personal|laboral|contacto|receptor|cliente)`

`DATA TYPE` `PA_L1_EMAIL` - Type

Accepts text (length 5 or more); any other type rules the domain out.

`VALUE FORMAT` `PA_L1_EMAIL` - Regex

PATTERN	STRENGTH	NOTES
<code>[A-Za-z0-9._%+'\-\-]+@[A-Za-z0-9.\-\-]+\.</code> <code>[A-Za-z]{2,24}</code>	100%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

The name and domain are mapped character by character and the top-level domain becomes `.test` , reserved for testing: a masked address never receives mail.

`PA_EMAIL` **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

yamileth.gonzalez@gmail.com

→ tificefk.yedsiquv@pgoeb.test

7.12 Phone

PA_L1_TELEFONO

ESSENTIAL

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Seven-digit landlines (223–4567) and eight-digit mobiles starting with 6 (6123–4567), with or without +507 . Panama has no area codes.

HOW IT IS FOUND

COLUMN NAME

PA_L1_TELEFONO – Path

85%

tel telefono[a-z0-9]* tel[eé]fono tel_?
(cel|casa|residencia|oficina|trabajo|movil|contacto|fijo|emergencia) celular[a-z0-9]* cel
movil m[oó]vil whatsapp wa fax phone[a-z0-9]* mobile cellphone n(o|um|ro)?
tel(efono)? telef tlf extension

VALUE FORMAT

PA_L1_TELEFONO – Regex

PATTERN

STRENGTH

NOTES

(\+?(?507\)?[\s-]?)?6\d{3}[\s-]?\d{4}

90%

(\+?(?507\)?[\s-]?)?[2-9]\d{2}[\s-]?\d{4}

70%

\+?\d[\d\s()\-]{6,15}

30%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

+507 , the separators and the first digit — the 6 of a mobile, the region of a landline — stay; the rest is masked digit by digit. A number not shaped like a Panamanian number is masked whole.

PA_TELEFONO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

+507 6123–4567

→ +507 6957–9910

223–4567

→ 246–9910

7.13 Address

PA_L1_DIRECCION

ESSENTIAL

COLUMN NAME

DATA TYPE

VALUE FORMAT

SECURE LOOKUP

WHAT IT IS

The address line as written in Panama, which rarely has a street number: street or road, building, floor and apartment, or barriada, street and house.

HOW IT IS FOUND

COLUMN NAME PA_L1_DIRECCION - Path

80% direccion[a-z]* (except when followed by list E5) direcci[oó]n dir domicilio[a-z]* calle(_?y_? casa)? avenida address (except when preceded by mac, ip, e?mail, web, url) street addr dir? (residencial|residencia|particular|comercial|laboral|envio|entrega|facturacion|cliente|paciente|trabajo) residencia lugar?(de_)?residencia home_address ubicacion? (domicilio|vivienda) senas punto_de_referencia

DATA TYPE PA_L1_DIRECCION - Type

Accepts text (length 6 or more); any other type rules the domain out.

VALUE FORMAT PA_L1_DIRECCION - Regex

PATTERN	STRENGTH	NOTES
(?i)(v[íí]a ave?\.? avenida calle c\. carretera paseo boulevard blvd\.? corredor camino)\s+.	70%	
(?i).*(edificio edif\.? torre ph)\s+.+(piso apto\.? apartamento oficina local)\s*\w+.	80%	
(?i).*(barriada urbanizaci[oó]n urb\. residencial sector manzana)\s+.+(casa lote)\s*(n[oó]\.?\s*)?\w+.	80%	

A value that matches none counts 20% against the domain.

HOW IT IS MASKED

It is replaced by one of 3,000 fictitious Panamanian addresses, with common streets, roads, barriadas and buildings in realistic proportions.

PA_DIRECCION SECURE LOOKUP

Replaces the value with one of the 3,000 values in pa-direcciones.txt, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Calle 50, Edificio Global Bank, Piso 21, Oficina 2104	→ Las Cumbres, Casa No. 242

7.14 Address detail

PA_L1_DIRECCION_COMPLEMENTO

ESSENTIAL COLUMN NAME CHARACTER MAPPING

WHAT IT IS

House, apartment, floor, building, tower or lot numbers in columns of their own.

HOW IT IS FOUND

COLUMN NAME PA_L1_DIRECCION_COMPLEMENTO - Path

70% n(o|um|ro|umero)_?casa casa?(num|no|numero) apto apartamento n(o|um|ro)_?apto piso edificio torre ph lote manzana local oficina_num house_number apartment apt unit_number floor

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

Casa 23

MASKED

→ Miya 82

7.15 Bank account

PA_L1_CUENTA_BANCARIA

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Checking, savings and payroll account numbers, in each bank's formats.

HOW IT IS FOUND

COLUMN NAME

PA_L1_CUENTA_BANCARIA - Path

85%

cuenta_?

(bancaria|banco|corriente|ahorro|ahorros|deposito|abono|pago|destino|origen|planilla|debito|ac
h|plazo_?fijo) cta_?(bancaria|banco|corriente|ahorro|planilla|ach) n(o|um|ro|umero)?_
(de_)?cuenta(_?(banco|bancaria|corriente|ahorro|ach))? account_?(number|no|num|nbr) bank_
account num_?cta no_?cta iban

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

04-72-01-123456-7

MASKED

→ 25-71-81-450676-9

7.16 Payment card

PA_L1_TARJETA

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Credit, debit and prepaid card numbers, including the Tarjeta Clave Social through which MIDES pays its programs.

HOW IT IS FOUND

COLUMN NAME

PA_L1_TARJETA - Path

85%

tarjeta(_?(credito|debito|cred|deb|num|no|numero|bancaria|clave|clave_?social|prepago))?

clave_?social n(o|um|ro|umero)?_?tarjeta pan card_?(number|no|num) credit_?card cc_
(number|num) tdc tdd num_?tdc plastico

VALUE FORMAT

PA_L1_TARJETA - Regex

PATTERN

STRENGTH

NOTES

\d{13,19}

90% LUHN check digit; after removing [\s\~]

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by another card number that still passes the Luhn check. Values not shaped like a card pass unchanged instead of failing the row.

PA_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PA_TARJETA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

EXAMPLES

VALUE	MASKED
4916 3131 2345 6780	→ 1374 9602 1069 5000

7.17 IP address

PA_L1_IP

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

IPv4 and IPv6 addresses of users and devices.

HOW IT IS FOUND

COLUMN NAME PA_L1_IP - Path

85%

ip ip_?

(address|addr|origen|destino|cliente|usuario|remota|publica|privada|acceso|conexion|login)

direccion_?ip dir_?ip ipv[46] remote_?addr client_?ip host_?ip

VALUE FORMAT PA_L1_IP - Regex

PATTERN

STRENGTH

NOTES

((25[0-5]|2[0-4]\d|1?\d?\d)\.){3}(25[0-5]|2[0-4]\d|1?\d?\d)

100%

[0-9A-Fa-f]{1,4}(:[0-9A-Fa-f]{0,4}){2,7}

90%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Each IPv4 octet is mapped to a number from 1 to 254; an IPv6 address is mapped as hexadecimal.

PA_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_HEX .

PA_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

PA_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
190.140.22.10	→ 164.230.40.16

7.18 Device identifier

PA_L1_DISPOSITIVO

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

IMEI, MAC address, advertising and session identifiers, equipment serial numbers.

HOW IT IS FOUND

COLUMN NAME PA_L1_DISPOSITIVO - Path

80% imei imsi iccid mac mac_address direccion_mac device_id id_dispositivo
dispositivo_id uuid_dispositivo advertising_id idfa gaid cookie(_id)? session_
id id_sesion serie_equipo n(o|u|m|r)o_serie_(equipo|celular|telefono)

VALUE FORMAT PA_L1_DISPOSITIVO - Regex

PATTERN	STRENGTH	NOTES
<code>([0-9A-Fa-f]{2}[:\-]){5}[0-9A-Fa-f]{2}</code>	100%	
<code>\d{15}</code>	80%	LUHN check digit

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Hex stays hex and digits stay digits, so the value keeps its format.

PA_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
3c:22:fb:9a:10:e4	→ 5e:83:bb:3c:31:f9

7.19 License plate

PA_L1_PLACA

COLUMN NAME VALUE FORMAT REGEX DECOMPOSE

WHAT IT IS

Plates in the format used since 2013, two letters and four digits (AB1234), the earlier six-digit ones, and category plates: M motorcycle, T taxi, B bus, G government, CD diplomatic corps, PR press and others.

HOW IT IS FOUND

COLUMN NAME PA_L1_PLACA - Path

85% placas?(?(\vehiculo|auto|circulacion|vehicular|num|no|numero))? n(o|u|m|r)o_placas?
matricula(?(\vehiculo|auto|vehicular))? license_plate plate(_number)? registro_
vehicular revisado

VALUE FORMAT PA_L1_PLACA - Regex

PATTERN	STRENGTH	NOTES
<code>[A-Z]{2}[\s\~]?d{4}</code>	80%	
<code>(ADM MB MI PR BC CD CC M T B G D E)[\s\~]?d{3,6}</code>	70%	
<code>\d{6}</code>	30%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Serial letters and digits are mapped; category letters stay, so a taxi plate is still a taxi plate.

PA_PLACA REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
AB1234	→	EX9707
T12345	→	T31564

7.20 Vehicle identification number

PA_L1_VIN

COLUMN NAME VALUE FORMAT CHARACTER MAPPING

WHAT IT IS

The VIN, and engine and chassis numbers.

HOW IT IS FOUND

COLUMN NAME PA_L1_VIN - Path

80% vin n(o|um|ro|umero)?(de_)?(serie|identificacion)?(vehicular|vehiculo) serie?
vehicul(o|ar) n(o|um|ro)?serie numero?serie n(o|um|ro)?motor chasis n(o|um|ro)?
chasis vin?number

VALUE FORMAT PA_L1_VIN - Regex

PATTERN	STRENGTH	NOTES
[A-HJ-NPR-Z0-9]{17}		70%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
3VWFE21C04M000001	→	7VVCU97M86J091175

7.21 Property number (finca)

PA_L1_FINCA

COLUMN NAME CHARACTER MAPPING

WHAT IT IS

A property's finca number, folio real or cadastral location code. The Public Registry can be searched by anyone: the finca leads to its owner.

HOW IT IS FOUND

COLUMN NAME PA_L1_FINCA - Path

85% finca n(o|um|ro|umero)?(de_)?finca folio_?real codigo_?(de_)?ubicacion numero_?catastral n(o|um|ro)?catastro catastro inmueble_?(num|no|numero|registro) registro_?(de_?la_)?propiedad

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
30162456	→	39020887

7.22 Contract or customer number

PA_L1_NUMERO_CONTRATO

COLUMN NAME **CHARACTER MAPPING**

WHAT IT IS

Contract, policy, loan, customer, employee, payroll or claim numbers, and the electricity service's NIS.

HOW IT IS FOUND

COLUMN NAME PA_L1_NUMERO_CONTRATO - Path

70% contrato n(o|um|ro|umero)?contrato p[ó]liza n(o|um|ro|umero)?poliza n(o|um|ro)?(de_)?(prestamo|credito|hipoteca) n(o|um|ro)?operacion n(o|um|ro|umero)?(de_)?socio n(o|um|ro|umero)?(de_)?cliente id_cliente cod(igo)?_cliente nis n(o|um|ro)?(de_)?medidor n(o|um|ro|umero)?(de_)?empleado num_emp no_emp n(o|um|ro)?planilla posicion_planilla n(o|um|ro)?sinistro n(o|um|ro)?(de_)?matricula n(o|um|ro)?expediente_(cliente|laboral|personal) folio_(contrato|poliza|credito|cliente)

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
POL-123456	→	MIF-194239

7.23 User name or social network

PA_L1_USUARIO_RRSS

COLUMN NAME **CHARACTER MAPPING**

WHAT IT IS

User names, aliases and social network profiles.

HOW IT IS FOUND

COLUMN NAME PA_L1_USUARIO_RRSS - Path

70% usuario username user_?name login nick(name)? alias cuenta_?usuario usr facebook
instagram twitter tiktok linkedin red_?social perfil_
(facebook|instagram|twitter|linkedin|tiktok) telegram handle

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

@yami_gonzalez

→

@denu_colwawem

7.24 Credential

PA_L1_CREDENCIAL

COLUMN NAME **REGEX DECOMPOSE**

WHAT IT IS

Passwords, PINs, password hashes, tokens and API keys.

HOW IT IS FOUND

COLUMN NAME PA_L1_CREDENCIAL - Path

90% password passwd pwd pass contrasen[aa] contraseña clave[a-z]* (except when followed by list E6)
pin nip hash_?password password_?hash secret token_?acceso access_?token refresh_
token api_?key otp respuesta_?secreta pregunta_?secreta

HOW IT IS MASKED

Every character becomes X . A credential has no reason to look real in a test environment.

PA_REDACTAR **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

Secreta#2024

→

XXXXXXXXXXXX

7.25 Geolocation

PA_L1_GEOLOCALIZACION

COLUMN NAME **VALUE FORMAT** **REGEX DECOMPOSE**

WHAT IT IS

Latitude and longitude, together or in separate columns. Panama lies between 7° and 10° north and 77° and 83° west.

HOW IT IS FOUND

COLUMN NAME	PA_L1_GEOLOCALIZACION - Path	
85%	<code>lat</code> <code>latitud</code> <code>latitude</code> <code>lon</code> <code>lng</code> <code>longitude</code> <code>coordenadas?</code> <code>coord_?[xy]</code> <code>geo_?</code> <code>(lat lon lng loc localizacion posicion)</code> <code>geolocalizacion</code> <code>ubicacion_?gps</code> <code>gps</code> <code>posicion_?</code> <code>gps</code>	
50%	<code>longitud</code> <code>long</code>	
VALUE FORMAT	PA_L1_GEOLOCALIZACION - Regex	
PATTERN	STRENGTH	NOTES
<code>[7-9]\.\d{3,}\s*,\s*-(7[7-9] 8[0-3])\.\d{3,}</code>	90%	
<code>[7-9]\.\d{3,}</code>	60%	
<code>-(7[7-9] 8[0-3])\.\d{3,}</code>	70%	
A value that matches none counts 30% against the domain.		

HOW IT IS MASKED

The integer part and first decimal stay — about 11 km — and the rest is masked: the place is generalized, not moved.

PA_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_CM_ALFANUM	CHARACTER MAPPING
Maps each character to another of its group (7 groups), keeping length and format.	

EXAMPLES

VALUE	MASKED
8.983333,-79.516667	→ 8.903230,-79.596978

7.26 Free text

PA_L1_TEXT0_LIBRE

COLUMN NAME	DATA TYPE	VALUE FORMAT	FREE TEXT REDACTION
-------------	-----------	--------------	---------------------

WHAT IT IS

Remarks, comments, complaints and notes where personal data is written without structure.

HOW IT IS FOUND

COLUMN NAME	PA_L1_TEXT0_LIBRE – Path
60%	observaciones? obs comentarios? notas? detalle descripcion? (caso queja reclamo incidente solicitud problema atencion hechos) texto?libre mensaje queja reclamo referencias? motivo?(texto detalle) narrativa relato resumen?caso remarks comments? notes? free?text memo bitacora
DATA TYPE	PA_L1_TEXT0_LIBRE – Type
Accepts text (length 30 or more); any other type rules the domain out.	

VALUE FORMAT PA_L1_TEXT0_LIBRE – Regex

PATTERN	STRENGTH	NOTES
<code>(([1-9] 1[0-3]) (AV PI)? E N PE) - \d{1,4} - \d{1,6}</code>	60%	anywhere in the value
<code>[\w.+\-]+@[\w\-\-]+\.[A-Za-z]{2,}</code>	70%	anywhere in the value
<code>(\+?507\s?)?6\d{3}-?\d{4}</code>	50%	anywhere in the value

A value that matches none does not count against the domain.

HOW IT IS MASKED

Free Text Redaction: cédulas, e-mails, mobiles, hyphenated landlines, card numbers, plates and IPs become [DAT0] , and the given names and surnames of the lists become [NOMBRE] .

PA_TEXT0_LIBRE FREE TEXT REDACTION

Replaces the text matching 7 patterns with [DAT0] , and the 1,094 terms of `pa-texto-nombres.txt` with [NOMBRE] .

EXAMPLES

VALUE		MASKED
Cliente González, cédula 8-442-445, correo y.gonzalez@gmail.com, tel 6123-4567.	→	Cliente [NOMBRE], cédula [DAT0], correo [DAT0], tel [DAT0].

Quasi-identifiers

Data that identifies no one on its own, but does in combination. Panama has 699 corregimientos, and 616 of them had fewer than 10,000 inhabitants in the 2023 census: a birth date, a sex and one of those corregimientos point at a handful of people. These attributes are generalized or shifted rather than replaced, and the data keeps its analytical value.

8.1 Birth date

PA_L2_FECHA_NACIMIENTO

ESSENTIAL

COLUMN NAME

DATA TYPE

DATE SHIFT

WHAT IT IS

The birth date, in date, timestamp or text columns.

HOW IT IS FOUND

COLUMN NAME

PA_L2_FECHA_NACIMIENTO – Path

90%

fecha_?(de_)?nac(imiento)? fec_?nac(imiento)? fch_?nac f_?nac(imiento|im)? fnac fechanac(imiento)? fecnac nacimiento dob birth_?date date_?of_?birth birthdate birthday cumplea(n|ñ)os

DATA TYPE

PA_L2_FECHA_NACIMIENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 60 days back or forward: it breaks the link to the person and keeps ages and cohorts almost intact.

PA_FECHA_NACIMIENTO

DATE SHIFT

Moves the date by -60 to 60 days.

EXAMPLES

VALUE		MASKED
1985-07-20	→	1985-08-05T00:00

8.2 Birth year

PA_L2_ANIO_NACIMIENTO

COLUMN NAME

DATA TYPE

REGEX DECOMPOSE

WHAT IT IS

The birth year alone.

HOW IT IS FOUND

COLUMN NAME

PA_L2_ANIO_NACIMIENTO – Path

90%

an(i)?o_?(de_)?nac(imiento)? año_?(de_)?nac(imiento)? birth_?year year_?of_?birth ano_?nacim

DATA TYPE

PA_L2_ANIO_NACIMIENTO – Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

The decade stays and the last digit is mapped: 1985 becomes another year of the eighties.

PA_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
1985	→	1980

8.3

Age

PA_L2_EDAD

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Ages in years, whole or with decimals.

HOW IT IS FOUND

COLUMN NAME PA_L2_EDAD – Path

85%

edad(_?

(actual|paciente|cliente|afiliado|asegurado|beneficiario|ingreso|anos|años|cumplida|jubilacion

))? age anos_?cumplidos rango_?(de_)?edad grupo_?(de_)?edad edad_?en_?anos

DATA TYPE PA_L2_EDAD – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

HOW IT IS MASKED

The tens digit stays and the units digit is mapped (37 becomes 3x). From 100 on, 90: extreme ages identify on their own.

PA_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE		MASKED
37	→	35
104	→	90

8.4

Life event dates

PA_L2_FECHA_EVENTO

COLUMN NAME DATA TYPE DATE SHIFT

WHAT IT IS

Dates of death, marriage, hiring and leaving a job, retirement, hospitalization, diagnosis, naturalization and other personal events.

HOW IT IS FOUND

COLUMN NAME PA_L2_FECHA_EVENTO – Path

80% fecha_?(de_)?(defuncion|fallecimiento|muerte|matrimonio|union|divorcio|ingreso|inicio_?labores|contratacion|contrato|egreso|despido|renuncia|jubilacion|pension|incapacidad|hospitalizacion|alta_?medica|diagnostico|atencion|consulta|parto|embarazo|detencion|sentencia|vacunacion|cirugia|reingreso|afiliacion|inscripcion_?css|naturalizacion|residencia) fec_?(defuncion|ingreso|egreso|contrato|atencion|diagnostico|incapacidad|jubilacion) fch_?(ingreso|egreso) death_?date hire_?date termination_?date date_?of_?death admission_?date discharge_?date

DATA TYPE PA_L2_FECHA_EVENTO – Type

Accepts date, TIMESTAMP or text (length 6 or more); any other type rules the domain out.

HOW IT IS MASKED

Shifted up to 30 days: less than the birth date, since they are often combined with it.

PA_FECHA_EVENTO **DATE SHIFT**

Moves the date by -30 to 30 days.

EXAMPLES

VALUE

MASKED

2021-03-15

→ 2021-03-05T00:00

8.5

Sex

PA_L2_SEX0

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Registered sex in any notation: F / M , H / M , Femenino , Hombre , Male .

HOW IT IS FOUND

COLUMN NAME PA_L2_SEX0 – Path

85% sexo(_?(biologico|registral|paciente|cliente|asegurado|trabajador))? sex g[eé]nero (except when followed by *identidad* , *social* , *autopercebido*) gender cod_?sexo

KNOWN VALUES PA_L2_SEX0 – List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-sexo.txt

11

50%

h, m, f, hombre, mujer, femenino

A value that matches none counts 50% against the domain.

HOW IT IS MASKED

Replaced within its own vocabulary: F / M stays F / M .

PA_SEX0 **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_SEX0_PALABRA **SECURE LOOKUP**

Replaces the value with one of the 2 values in pa-sexo-palabra.txt , chosen by a keyed hash.

PA_SEX0_MUJER_HOMBRE **SECURE LOOKUP**

Replaces the value with one of the 2 values in pa-sexo-mujer-hombre.txt , chosen by a keyed hash.

PA_SEX0_INGLES SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-ingles.txt`, chosen by a keyed hash.

PA_SEX0_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-f-m.txt`, chosen by a keyed hash.

PA_SEX0_H_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-h-m.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F	→	M
Masculino	→	Femenino

8.6 District

PA_L2_DISTRITO

COLUMN NAME

KNOWN VALUES

DATA CLEANSING

WHAT IT IS

The district name. Panama has 81 inhabited districts — the Guna Yala comarca is not divided into districts —, and 47 had fewer than 20,000 inhabitants in 2023.

HOW IT IS FOUND

COLUMN NAME

PA_L2_DISTRITO - Path

85%

`distrito[a-z]* nom(bre)?_?(dist|distrito) dist municipio mun distrito_?
(residencia|nacimiento|domicilio)`

KNOWN VALUES

PA_L2_DISTRITO - List

LIST

VALUES

STRENGTH

FIRST VALUES

`pa-detectar-
distritos.txt`

111

80%

Bocas del Toro, Changuinola, Chiriquí Grande, Chiriqui Grande, Almirante,
Aguadulce

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A district under 20,000 inhabitants becomes the nearest district of its province or comarca with at least 20,000; larger ones stay. See *Where people live*.

PA_DISTRITO DATA CLEANSING

Replaces each value found in `pa-distrito-generalizado.txt` (234 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
Santa Isabel	→	Colón
David	→	David

COLUMN NAME KNOWN VALUES DATA CLEANSING

WHAT IT IS

The corregimiento name, the smallest division with a census population: 699 of them, averaging 5,800 inhabitants.

HOW IT IS FOUND

COLUMN NAME

PA_L2_CORREGIMIENTO – Path

85%

corregimiento[a-z]* nom(bre)?_?(correg|corregimiento) correg corr corregimiento_?

(residencia|nacimiento|domicilio)

50%

ciudad[a-z]* city town poblacion

KNOWN VALUES

PA_L2_CORREGIMIENTO – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-corregimientos.txt	814	80%	Bocas del Toro, Bastimentos, Punta Laurel, Tierra Oscura, Bocas del Drago, San Cristóbal

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A corregimiento under 10,000 inhabitants becomes the nearest corregimiento of its province or comarca with at least 10,000, or of the country if the province has none. Large ones stay. See *Where people live*.

PA_CORREGIMIENTO DATA CLEANSING

Replaces each value found in pa-corregimiento-generalizado.txt (1,760 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE	MASKED
Cerro Punta	→ La Concepción
Tocumen	→ Tocumen

COLUMN NAME DATA TYPE KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

INEC geographic codes: six digits for a corregimiento (040612), four for a district (0306), also without the leading zero.

HOW IT IS FOUND

COLUMN NAME

PA_L2_CODIGO_LUGAR – Path

85%

cod(igo)?_?(correg|corregimiento|dist|distrito|lugar(_?poblado)?|inec|geografico) cve_?

(correg|dist|lugar) id_?(correg|corregimiento|distrito) correg_?(cod|id) distrito_?

(cod|id) codigo_?censal

DATA TYPE

PA_L2_CODIGO_LUGAR – Type

Accepts number (length up to 6) or text (length up to 6); any other type rules the domain out.

KNOWN VALUES PA_L2_CODIGO_LUGAR – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-codigo-lugar.txt	1,405	40%	010101, 10101, 010102, 10102, 010104, 10104

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

By the same rule as the names: a small corregimiento or district becomes the code of the large one it maps to.

PA_CODIGO_LUGAR **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_CODIGO_CORREGIMIENTO **DATA CLEANSING**

Replaces each value found in pa-codigo-corregimiento-generalizado.txt (1,331 pairs) by its counterpart; values not in the file stay as they are.

PA_CODIGO_DISTRITO **DATA CLEANSING**

Replaces each value found in pa-codigo-distrito-generalizado.txt (148 pairs) by its counterpart; values not in the file stay as they are.

EXAMPLES

VALUE		MASKED
040612	→	040612
0306	→	0301

8.9 Populated place

PA_L2_LUGAR_POBLADO

COLUMN NAME **SECURE LOOKUP**

WHAT IT IS

Populated places, hamlets and communities, and the place of birth.

HOW IT IS FOUND

COLUMN NAME PA_L2_LUGAR_POBLADO – Path

80% lugar_?poblado nom(bre)?_?lugar_?poblado poblado comunidad (except when followed by *indigena*, *religiosa*) caserio localidad lugar_?(de_)?nacimiento pueblo_?(residencia|origen)

HOW IT IS MASKED

Replaced by the name of a corregimiento of more than 20,000 inhabitants: still a place, no longer theirs.

PA_LUGAR_POBLADO **SECURE LOOKUP**

Replaces the value with one of the 52 values in pa-lugares-poblados.txt, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Los Llanos de Ocú	→	El Coco

8.10 Barriada or neighborhood

PA_L2_BARRIADA

COLUMN NAME **SECURE LOOKUP**

WHAT IT IS

Barriadas, urbanizaciones, residential developments and sectors.

HOW IT IS FOUND

COLUMN NAME PA_L2_BARRIADA – Path

85% `barriada[a-z]*` `barrio` `urbanizacion` `urb` `residencial` `sector` `nom(bre)?_?`
`(barriada|urbanizacion|barrio)` `condominio` `ph?nombre` `nombre?ph`

HOW IT IS MASKED

Replaced by one of 75 common *barriadas* and neighborhoods.

PA_BARRIADA **SECURE LOOKUP**

Replaces the value with one of the 75 values in `pa-barriadas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Villa Guadalupe

→

Las Acacias

8.11 Postal code

PA_L2_CODIGO_POSTAL

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Postal codes of Correos y Telégrafos, little used in Panama. The first two digits give the province.

HOW IT IS FOUND

COLUMN NAME PA_L2_CODIGO_POSTAL – Path

85% `cod(igo)?_?postal` `zona?postal` `cp` `zip(_?code)?` `postal?code` `codpostal` `apartado?`
`postal`

VALUE FORMAT PA_L2_CODIGO_POSTAL – Regex

PATTERN

STRENGTH

NOTES

`\d{4,5}`

20%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

The first two digits stay and the rest becomes zeros.

PA_CODIGO_POSTAL **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_ALFANUM.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

0801

→

0800

8.12 Nationality

PA_L2_NACIONALIDAD

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

Nationality or country of birth. In a Panamanian database, an uncommon nationality identifies.

HOW IT IS FOUND

COLUMN NAME

PA_L2_NACIONALIDAD – Path

80%

nacionalidadnationalitypais_?(de_)?(nacimiento|origen)ciudadaniacitizenshipcountry_of_birthextranjero

KNOWN VALUES

PA_L2_NACIONALIDAD – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-nacionalidades.txt	63	70%	Panameña, Panamena, Colombiana, Venezolana, Nicaragüense, Nicaraguense

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by one of the 20 most common nationalities in Panama.

PA_NACIONALIDAD **SECURE LOOKUP**
Replaces the value with one of the 20 values in pa-nacionalidades.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Venezolana	→ Estadounidense

8.13 Marital status

PA_L2_ESTADO_CIVIL

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

Marital status, in the census categories: in a union, married, single, separated from a marriage, separated from a union, divorced, widowed.

HOW IT IS FOUND

COLUMN NAME

PA_L2_ESTADO_CIVIL – Path

85%

estado_civiledo_civilest_civilcivilestado_conyugal situacion_conyugalmarital_statuscivil_status

KNOWN VALUES

PA_L2_ESTADO_CIVIL – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-estado-civil.txt	18	70%	soltero, soltera, casado, casada, unido, unida

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another of the seven categories.

PA_ESTADO_CIVIL **SECURE LOOKUP**
Replaces the value with one of the 7 values in pa-estado-civil.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Unida	→ Casado

8.14 Occupation

PA_L2_OCUPACION

COLUMN NAME SECURE LOOKUP

WHAT IT IS

Occupation, profession or job title. A rare title (Gerente Regional de Operaciones Bocas del Toro) is unique in the database.

HOW IT IS FOUND

COLUMN NAME PA_L2_OCUPACION – Path

80% ocupaci[oó]n profes[ioó]n oficio puesto(_?(trabajo|laboral|actual))? (except when followed by votacion , venta , trabajo_?id) cargo (except when followed by list E7) job_?title occupation actividad_?(laboral|economica_?persona) categoria_?laboral

HOW IT IS MASKED

Replaced by one of 40 common occupations.

PA_OCUPACION SECURE LOOKUP

Replaces the value with one of the 40 values in pa-ocupaciones.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Gerente Regional de Operaciones Bocas del Toro	→	Bodeguero

8.15 Employer

PA_L2_EMPLEADOR

COLUMN NAME SECURE LOOKUP

WHAT IT IS

The name of the employer (patrono) or workplace.

HOW IT IS FOUND

COLUMN NAME PA_L2_EMPLEADOR – Path

80% empleador patrono nombre_?patrono razon_?social_?(patrono|empleador) empresa_?(empleadora|trabajo|laboral) lugar_?(de_?)?trabajo centro_?(de_?)?trabajo employer institucion_?(donde_?)?labora entidad_?(donde_?)?labora

HOW IT IS MASKED

Replaced by one of 25 fictitious employers.

PA_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 25 values in pa-empleadores.txt , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Autoridad del Canal de Panamá	→	Ferretería EL Constructor, S.A.

8.16 Schooling

PA_L2_ESCOLARIDAD

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

Education level in the terms of the Panamanian system: primaria, premedia, media, vocational, higher education.

HOW IT IS FOUND

COLUMN NAME

PA_L2_ESCOLARIDAD - Path

80%

escolaridad nivel?(educativo|educacional|escolar|de?estudios|academico|de?instruccion)
grado?(de?)?(estudios|instruccion) grado?academico ultimo?grado(_?aprobado)?
education?level anos?(de?)?escolaridad

KNOWN VALUES

PA_L2_ESCOLARIDAD - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-escolaridad.txt

28

60%

Ningún grado, Ningun grado, Primaria incompleta, Primaria completa, Premedia incompleta, Premedia completa

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another of the 13 levels.

PA_ESCOLARIDAD SECURE LOOKUP

Replaces the value with one of the 13 values in pa-escolaridad.txt, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Premedia completa

→

Premedia incompleta

8.17 School

PA_L2_ESCUELA

COLUMN NAME

SECURE LOOKUP

WHAT IT IS

The name of the school, colegio, basic education center or university.

HOW IT IS FOUND

COLUMN NAME

PA_L2_ESCUELA - Path

75%

escuela colegio plantel nombre?(escuela|plantel|colegio|centro?educativo|institucion?
educativa) centro?educativo institucion?educativa universidad instituto school(_?
name)? cebg centro?basico

HOW IT IS MASKED

Replaced by one of 15 fictitious schools.

PA_ESCUELA SECURE LOOKUP

Replaces the value with one of the 15 values in pa-escuelas.txt, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Instituto Nacional de Panamá

→ Universidad Tecnológica del Pacífico

8.18 Children and dependents

PA_L2_DEPENDIENTES

COLUMN NAME DATA TYPE REGEX DECOMPOSE

WHAT IT IS

Number of children, dependents or household members.

HOW IT IS FOUND

COLUMN NAME PA_L2_DEPENDIENTES - Path

80% dependientes n(o|um|ro)?_?dependientes num_?hijos n(o|um|ro)?_?hijos cant_?hijos hijos hijos_?nacidos_?vivos integrantes_?(del_)?(hogar|familia) n(o|um|ro)_?integrantes tamano_?(del_)?hogar personas_?(en_)?(el_)?hogar

DATA TYPE PA_L2_DEPENDIENTES - Type

Accepts number (length up to 4) or text (length up to 4); any other type rules the domain out.

HOW IT IS MASKED

Capped at 5: large households are rare, and a rare count identifies.

PA_CANTIDAD_ACOTADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE

MASKED

9

→ 5

Sensitive data

The categories of article 4, section 11 — ethnic origin, health, genetic and biometric data, beliefs, political opinions, union membership, sexual orientation — and, because the list is not exhaustive, gender identity and sexual life. Categorical columns keep their shape: a yes/no flag stays a flag, a numeric code gets other digits and any other value is replaced by one from its category's list. Where the values share no vocabulary, they are suppressed to Sin información .

9.1 Ethnic origin

PA_L3_ETNIA

COLUMN NAME KNOWN VALUES REGEX DECOMPOSE

WHAT IT IS

Belonging to an indigenous people — Ngäbe, Guna, Emberá, Buglé, Wounaan, Naso Tjërdi, Bokota, Bri Bri — or Afro-descendant self-identification — Afro-Panamanian, moreno, black, Afro-colonial, Afro-Antillean —, in the terms of the 2023 census, often as a yes/no flag.

HOW IT IS FOUND

COLUMN NAME PA_L3_ETNIA – Path

90%

etnia [eé]tnico origen_[eé]tnico grupo_[eé]tnico pueblo_?(indigena|originario) grupo_?indigena indigena ind[ií]lgena afro(panameno|descendiente|panameño)? grupo_?afro(descendiente)? autoidentificacion(_?(etnica|indigena|afro))? se_?considera_?(indigena|afro|afrodescendiente) raza race ethnicity ethnic_?group comunidad_?indigena pertenencia_?(etnica|indigena)

KNOWN VALUES PA_L3_ETNIA – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-pueblos.txt	52	90%	Guna, Ngäbe, Ngabe, Buglé, Bugle, Naso Tjërdi
pa-detectar-bandera.txt	8	30%	s, n, sí, sí, no, y

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

A flag stays a flag in its vocabulary, a code gets other digits and any other value is replaced by one from the census list.

PA_PUEBLO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_PUEBLO_LISTA .

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_PUEBLO_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `pa-pueblos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Teribe	→	Moreno
S	→	S

9.2 Indigenous language

PA_L3 LENGUA_INDIGENA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The indigenous language the person speaks: Ngäbere, Buglere, Guna, Emberá, Wounmeu, Naso Tjërdi, Bribri.

HOW IT IS FOUND

COLUMN NAME

PA_L3 LENGUA_INDIGENA - Path

90%

`lengu(?:indigena|materna|originaria)? habla?(lengu(?:indigena|originaria) idioma?(indigena|materno|originario) lengua_?que?habla dialecto native_?language`

KNOWN VALUES

PA_L3 LENGUA_INDIGENA - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-lenguas.txt

32

80%

Ngäbere, Ngabere, Buglere, Guna, Emberá, Embera

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Like ethnic origin: flags and codes keep their shape, names are replaced by another language from the list.

PA LENGUA_INDIGENA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA LENGUA_INDIGENA LISTA

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA LENGUA_INDIGENA LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-lenguas.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Ngäbere

→ No habla lengua indígena

9.3 Diagnosis

PA_L3_SALUD_DIAGNOSTICO

COLUMN NAME

VALUE FORMAT

REGEX DECOMPOSE

WHAT IT IS

Diagnoses, as ICD-10 codes (F32.9 , B20) or text, and medical history, allergies and chronic conditions.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_DIAGNOSTICO - Path

85%

diagn[oó]stico[a-z]* diag dx cie_?10 cie_?11 cod(igo)?_?cie cie icd_?10 icd
patolog[ií]a[a-z]* padecimiento[a-z]* enfermedad(es)?[a-z]* morbilidad condicion_
(medica|de_salud|cronica) problema_(de_)?salud impresion_diagnostica causa_
(incapacidad|muerte|defuncion|hospitalizacion|consulta) motivo_(de_)?
(consulta|hospitalizacion|incapacidad) comorbilidad(es)? antecedentes_
(medicos|patologicos|personales_patologicos|heredofamiliares|clinicos) alergias?
enfermedad_preexistente preexistencias? enfermedad_cronica riesgo_profesional

VALUE FORMAT

PA_L3_SALUD_DIAGNOSTICO - Regex

PATTERN

STRENGTH

NOTES

[A-TV-Za-tv-z]\d{2}(\.\d{1,2}|\d)?

50%

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

An ICD-10 code is replaced by another common code of the same format, with or without a decimal; text, by a common diagnosis. The real diagnosis does not survive.

PA_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PA_DIAGNOSTICO_TEXT0 .

PA_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in pa-cie10-decimal.txt , chosen by a keyed hash.

PA_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in pa-cie10.txt , chosen by a keyed hash.

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_DIAGNOSTICO_TEXTO **SECURE LOOKUP**

Replaces the value with one of the 30 values in `pa-diagnosticos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
F32.9	→	J02.9
Depresión mayor	→	Tendinitis

9.4 Medication

PA_L3_SALUD_MEDICAMENTO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Medications and active ingredients. Some reveal the diagnosis: an antiretroviral, an antipsychotic.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_MEDICAMENTO - Path

85%

medicamentos? f[aá]rmascos? sustancia_?activa principio_?activo prescripci[oó]n
receta(_?medica)? tratamiento(_?farmacologico)? medication drug(_?name)? posologia
lista_?oficial_?medicamentos

KNOWN VALUES

PA_L3_SALUD_MEDICAMENTO - List

LIST

VALUES STRENGTH FIRST VALUES

pa-detectar- medicamentos.txt	58	70%	Acetaminofén, Acetaminofen, Ibuprofeno, Naproxeno, Diclofenaco, Ketorolaco
----------------------------------	----	-----	--

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Replaced by one of 26 common medications.

PA_MEDICAMENTO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_MEDICAMENTO_LISTA .

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_MEDICAMENTO_LISTA **SECURE LOOKUP**

Replaces the value with one of the 26 values in `pa-medicamentos.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Sertralina 50 mg

→ Amoxicilina

9.5 Clinical identifier

PA_L3_SALUD_IDENTIFICADOR

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Medical record, file, admission, prescription or sick-leave number.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_IDENTIFICADOR - Path

85%

expediente_?clinico n(o|um|ro|umero)_?expediente(_?clinico)? historia_?clinica
n(o|um|ro)_?historia nhc n(o|um|ro)_?episodio n(o|um|ro)_?(de_)?admision folio_
(atencion|urgencias|receta|incapacidad) n(o|um|ro)_?incapacidad certificado_?(de_)?
incapacidad n(o|um|ro)_?(de_)?cama n(o|um|ro)_?interconsulta n(o|um|ro)_?(de_)?receta
n(o|um|ro)_?biopsia medical_?record_?number mrn

55%

expediente episodio cama

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

HC-2024-00123

→ TW-3389-01317

9.6 Clinical text

PA_L3_SALUD_TEXTO_CLINICO

COLUMN NAME

DATA TYPE

FREE TEXT REDACTION

WHAT IT IS

Progress notes, medical history, discharge summaries and clinical summaries.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_TEXTO_CLINICO - Path

85%

nota_?(medica|de_?evolucion|evolucion|de_?ingreso|ingreso|de_?egreso|egreso|de_?
urgencias|urgencias|postoperatoria|preoperatoria|de_?enfermeria|clinica)s? notas_?
clinicas? enfermedad_?actual anamnesis examen_?fisico evoluci[oó]n(_?(medica|clinica))?
epicrisis indicaciones(_?medicas)? resumen_?clinico resumen_?(de_)?egreso historia_
(de_?la_)?enfermedad plan_?(de_)?tratamiento pronostico hallazgos interpretacion
clinical_?notes? progress_?notes?

DATA TYPE

PA_L3_SALUD_TEXTO_CLINICO - Type

Accepts text (length 30 or more); any other type rules the domain out.

HOW IT IS MASKED

Free Text Redaction, as for free text: cédulas, contact details and names are redacted. The clinical content stays; if it must go too, assign the column to a suppression domain.

PA_TEXTO_LIBRE **FREE TEXT REDACTION**

Replaces the text matching 7 patterns with [DATO], and the 1,094 terms of pa-texto-nombres.txt with [NOMBRE].

EXAMPLES

VALUE

MASKED

Paciente Batista, cédula 8-442-445, acude por cefalea. → Paciente [NOMBRE], cédula [DATO], acude por cefalea.

9.7 Test result

PA_L3_SALUD_RESULTADO_EXAMEN

COLUMN NAME

CHARACTER MAPPING

WHAT IT IS

Laboratory and imaging results: HIV, viral load, glucose, pregnancy or drug tests.

HOW IT IS FOUND

COLUMN NAME PA_L3_SALUD_RESULTADO_EXAMEN - Path

80%

resultado_?

(estudio|examen|laboratorio|lab|prueba|pcr|antigeno|biopsia|imagen|vih|covid|papanicolaou|mamografia) examenes?_(de_)?laboratorio vih hiv carga_?viral cd4 glucosa glicemia hemoglobina(_?glicosilada)? hba1c colesterol trigliceridos presion_?arterial imc prueba_?(covid|embarazo|vih|antidoping|toxicologica) antidoping lab_?result

50%

resultado examen

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE

MASKED

Reactivo → Quilqaju

9.8 Blood group

PA_L3_SALUD_GRUPO_SANGUINEO

COLUMN NAME

KNOWN VALUES

SECURE LOOKUP

WHAT IT IS

ABO group and Rh factor.

HOW IT IS FOUND

COLUMN NAME PA_L3_SALUD_GRUPO_SANGUINEO - Path

90%

grupo_?sangu[ií]neo grupo_?sang tipo_?(de_)?sangre tipo_?sanguineo factor_?rh rh blood_?type blood_?group

KNOWN VALUES PA_L3_SALUD_GRUPO_SANGUINEO - List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-grupo-sanguineo.txt	26	90%	A+, A-, B+, B-, AB+, AB-

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another of the eight groups.

PA_GRUPO_SANGUINEO **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pa-grupo-sanguineo.txt`, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
0+	→ A-

9.9 Disability

PA_L3_SALUD_DISCAPACIDAD

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Type or degree of disability, in the types SENADIS certifies: physical, visual, hearing, intellectual, psychosocial, visceral.

HOW IT IS FOUND

COLUMN NAME PA_L3_SALUD_DISCAPACIDAD - Path

90% `discapacidad[a-z]* tipo_(de_)?discapacidad grado_(de_)?discapacidad certificado_(de_)?discapacidad carne_senadis senadis persona_con_discapacidad pcd invalidez porcentaje_(de_)?invalidez pension_(de_)?invalidez limitacion(_?(fisica|actividad))? disability dependencia_funcional nee necesidades_educativas_especiales`

KNOWN VALUES PA_L3_SALUD_DISCAPACIDAD - List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-discapacidad.txt	26	60%	Física, Física, Visual, Auditiva, Intelectual, Psicosocial

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the type is replaced by another from the list.

PA_DISCAPACIDAD **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_DISCAPACIDAD_LISTA.

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_DISCAPACIDAD_LISTA **SECURE LOOKUP**

Replaces the value with one of the 8 values in `pa-discapacidad.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Discapacidad psicosocial severa

→ **Auditiva**

9.10 Sexual and reproductive health

PA_L3_SALUD_SEXUAL_REPRODUCTIVA

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Pregnancy, weeks of gestation, abortions, contraception, fertility, sexually transmitted infections.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_SEXUAL_REPRODUCTIVA - Path

90%

`embarazo` `embarazada` `gestaci[oó]n` `semanas_?(de_)?gestacion` `edad_?gestacional` `fum`
`fecha_?ultima_?menstruacion` `aborto` `anticonceptivos?` `metodo_?(de_)?planificacion(_?`
`familiar)?` `planificacion_?familiar` `fertilidad` `infertilidad` `reproduccion_?asistida`
`gestas` `partos` `cesareas?` `puerperio` `its` `ets` `infeccion_?(de_)?transmision_?sexual`
`prep` `pregnan[a-z]*` `contracepti[a-z]*`

HOW IT IS MASKED

Flags keep their shape; any other value becomes `Sin información` .

PA_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR` .

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `keep`.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_SUPRIMIR **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Embarazo de 12 semanas

→

Sin información

S

→

S

9.11 Mental health and addiction

PA_L3_SALUD_MENTAL_ADICCIONES

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

Mental disorders, suicide attempts, substance use and addiction treatment.

HOW IT IS FOUND

COLUMN NAME

PA_L3_SALUD_MENTAL_ADICCIONES - Path

90%

salud_?mental psiquiatri[a-z]* psicologi[a-z]* trastorno[a-z]* depresi[oón]n ansiedad
suicid[a-z]* autolesi[oón]n adicci[oón]n(es)? consumo_?(de_?)?
(drogas|alcohol|sustancias|tabaco) alcoholismo farmacodependencia drogadiccion
toxicomania dependencia_?(alcohol|drogas|sustancias) tratamiento_?
(adicciones|drogas|alcohol) rehabilitacion_?(adicciones|drogas) mental_?health substance_?
(use|abuse)

HOW IT IS MASKED

Suppressed to Sin información .

PA_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_SUPRIMIR .

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in pa-sin-informacion.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Trastorno bipolar

→

Sin información

9.12 Biometric data

PA_L3_BIOMETRICO

COLUMN NAME **DATA TYPE** **REGEX DECOMPOSE**

WHAT IT IS

Fingerprint, face, iris or voice templates stored as text (hash, base64). The law names them sensitive and Decree 285 defines them.

HOW IT IS FOUND

COLUMN NAME PA_L3_BIOMETRICO - Path

90% huella(_?(dactilar|digital))? huellas dactilar biometri[a-z]* template_?
(facial|biometrico|huella) rostro reconocimiento_?facial facial_?(template|id) iris
retina voz_?(huella|biometrica) firma_?(digitalizada|biometrica|autografa_?digital)
fingerprint face_?(id|template|encoding) palm_?print minucias

DATA TYPE PA_L3_BIOMETRICO - Type

Accepts text; any other type rules the domain out.

HOW IT IS MASKED

Every character becomes **X** . Binary columns are out of reach: see *Limits*.

PA_REDACTAR **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE	MASKED
A1B2C3D4E5F6	→ XXXXXXXXXXXX

9.13 Genetic data

PA_L3_INFORMACION_GENETICA

COLUMN NAME **REGEX DECOMPOSE**

WHAT IT IS

Genetic profiles, sequences, mutations, karyotypes, paternity tests.

HOW IT IS FOUND

COLUMN NAME PA_L3_INFORMACION_GENETICA - Path

90% adn dna genoma genotipo genetic[a-z]* gen[eé]tic[ao] informacion_genetica
secuencia_genetica marcador_genetico perfil_(genetico|biologico) mutacion(_?
genetica)? snp brca haplotipo cariotipo tamizaje_(genetico|neonatal|metabolico)
prueba_(de_)?paternidad

HOW IT IS MASKED

Every character becomes **X** .

PA_REDACTAR **REGEX DECOMPOSE**

Matches one pattern and gives each part of the value its own action; when none matches: keep.

EXAMPLES

VALUE	MASKED
BRCA1 positivo	→ XXXXXXXXXXXX

9.14 Sexual orientation

PA_L3_ORIENTACION_SEXUAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Sexual orientation or preference, which the law names explicitly.

HOW IT IS FOUND

COLUMN NAME

PA_L3_ORIENTACION_SEXUAL - Path

95%

orientaci[oó]n?sexual

preferencia?sexual

sexual?(orientation|preference)

orient_?

sexual

pref?sexual

KNOWN VALUES

PA_L3_ORIENTACION_SEXUAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-orientacion-
sexual.txt

17

80%

Heterosexual, Homosexual, Lesbiana, Gay, Bisexual,
Pansexual

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the value is replaced by another from the list.

PA_ORIENTACION_SEXUAL

REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_ORIENTACION_SEXUAL_LISTA .

PA_BANDERA

REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO

SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE

SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

PA_BANDERA_S_N

SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

PA_BANDERA_Y_N

SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

PA_BANDERA_0_1

SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

PA_CM_ALFANUM

CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_ORIENTACION_SEXUAL_LISTA

SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-orientacion-sexual.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Bisexual

→

Lesbiana

9.15 Gender identity

PA_L3_IDENTIDAD_GENERO

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Gender identity, name changes for identity, pronouns.

HOW IT IS FOUND

COLUMN NAME

PA_L3_IDENTIDAD_GENERO - Path

95%

identidad_(de_)?g[eé]nero

gender_identity

g[eé]nero?(identidad|social|autopercibido)

transg[eé]nero

transsexual

persona_?trans

cambio_(de_)?(sexo|genero|nombre_?por_?

genero)

pronombres?

KNOWN VALUES

PA_L3_IDENTIDAD_GENERO - List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-identidad-genero.txt	17	70%	Mujer, Hombre, Mujer trans, Hombre trans, No binario, Género fluido

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the value is replaced by another from the list.

PA_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_IDENTIDAD_GENERO_LISTA .

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in pa-identidad-genero.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Hombre trans	→ No binario

9.16 Sexual life

PA_L3_VIDA_SEXUAL

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Sexual activity, behavior and history.

HOW IT IS FOUND

COLUMN NAME	PA_L3_VIDA_SEXUAL – Path
90%	vida_?sexual conducta_?sexual actividad_?sexual parejas?_?sexuales? practicas_?sexuales inicio_?(de_?)(la_?)?vida_?sexual sexual_?(activity behavior history) historia_?sexual comportamiento_?sexual

HOW IT IS MASKED

Suppressed to Sin información .

PA_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_SUPRIMIR .

PA_BANDERA REGEX DECOMPOSE
Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.
PA_BANDERA_SI_NO SECURE LOOKUP
Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.
PA_BANDERA_TRUE_FALSE SECURE LOOKUP
Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.
PA_BANDERA_S_N SECURE LOOKUP
Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.
PA_BANDERA_Y_N SECURE LOOKUP
Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.
PA_BANDERA_0_1 SECURE LOOKUP
Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.
PA_CM_ALFANUM CHARACTER MAPPING
Maps each character to another of its group (7 groups), keeping length and format.
PA_SUPRIMIR SECURE LOOKUP
Replaces the value with one of the 1 values in pa-sin-informacion.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Activa	→ Sin información

9.17 Religion

PA_L3_RELIGION

COLUMN NAME	KNOWN VALUES	REGEX DECOMPOSE
-------------	--------------	-----------------

WHAT IT IS

Religion or denomination: Catholic, Evangelical, Adventist, Jehovah's Witness, Bahá'í, Jewish, Muslim, and the indigenous religions Mamatata and Ibeorgun.

HOW IT IS FOUND

COLUMN NAME	PA_L3_RELIGION – Path
95%	religi[oó]n[a-z]* credo culto iglesia creencia_?religiosa confesion_?religiosa denominacion_?religiosa religious_?affiliation comunidad_?religiosa congregacion

KNOWN VALUES PA_L3_RELIGION – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-religiones.txt	61	90%	Católica, Catolica, Evangélica, Evangelica, Adventista del Séptimo Día, Adventista del Septimo Dia

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the value is replaced by another from the list.

PA_RELIGION **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_RELIGION_LISTA .**PA_BANDERA** **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_RELIGION_LISTA **SECURE LOOKUP**

Replaces the value with one of the 14 values in `pa-religiones.txt` , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Episcopal	→ Musúľmana

9.18 Philosophical and moral beliefs

PA_L3_CREENCIA_FILOSOFICA_MORAL

COLUMN NAME **REGEX DECOMPOSE****WHAT IT IS**

Personal convictions, conscientious objection, ethical positions.

HOW IT IS FOUND**COLUMN NAME** PA_L3_CREENCIA_FILOSOFICA_MORAL – Path

90% `creencias?_(filosoficas?|morales?|personales)` `convicci[oó]n(es)?(_?(filosoficas?|morales?`
`|eticas?|personales))?` `objecion?(de_)?conciencia` `postura?(etica|moral)` `valores_?`
`personales` `filosofia?(de_)?vida` `cosmovision`

HOW IT IS MASKED

Suppressed to Sin información .

PA_CATEGORIA_SUPRIMIDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR`.

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

`PA_BANDERA_SI_NO` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_SUPRIMIR` **SECURE LOOKUP**

Replaces the value with one of the 1 values in `pa-sin-informacion.txt`, chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Rechaza transfusiones	→ Sin información

9.19 Political opinion

`PA_L3_OPINION_POLITICA`

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Ideology, electoral preference, voting intention.

HOW IT IS FOUND

COLUMN NAME `PA_L3_OPINION_POLITICA` - Path

90% `opini[oó]n(es)?_pol[ií]ticas?` `ideolog[ií]a(_?politica)?` `posicion_?politica` `postura_?`
`politica` `tendencia_?politica` `orientacion_?politica` `preferencia_?`
`(politica|electoral|partidista)` `intencion_?(de_)?voto` `simpatia_?(politica|partidista)`
`political_?(view|orientation|opinion)`

KNOWN VALUES `PA_L3_OPINION_POLITICA` - List

LIST	VALUES	STRENGTH	FIRST VALUES
<code>pa-detectar-opiniones-politicas.txt</code>	14	70%	Izquierda, Centro izquierda, Centro, Centro derecha, Derecha, Sin preferencia

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by another position on a generic scale.

`PA_OPINION_POLITICA` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_OPINION_POLITICA_LISTA`.

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_OPINION_POLITICA_LISTA **SECURE LOOKUP**

Replaces the value with one of the 7 values in `pa-opiniones-politicas.txt` , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Centro derecha	→ Centro

9.20 Political affiliation

PA_L3_AFILIACION_POLITICA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

The party a person is registered with or supports. The Electoral Tribunal counted more than 1.5 million party members in 2026.

HOW IT IS FOUND

COLUMN NAME

PA_L3_AFILIACION_POLITICA - Path

95%

partido(?:pol[í]tico)? militancia militante adherente inscrito?partido
afiliaci[oó]n?(politica|partidista|partido) afiliado?(a_)?partido padron?(de_)?
(militantes|adherentes) political?party party?affiliation alianza?electoral
simpatizante libre?postulacion

KNOWN VALUES

PA_L3_AFILIACION_POLITICA - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-partidos.txt	35	80%	Partido Revolucionario Democrático, Partido Revolucionario Democratico, Realizando Metas, Cambio Democrático, Cambio Democratico, Partido Panameñista
--------------------------	----	-----	---

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Replaced by one of the legally constituted parties — PRD, Realizando Metas, Cambio Democrático, Panameñista, MOLIRENA, MOCA, FAD, Alianza, Partido Popular — or by independent candidacy (*libre postulación*) or no party.

PA_PARTIDO_POLITICO **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_PARTIDO_POLITICO_LISTA .

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_PARTIDO_POLITICO_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in pa-partidos.txt , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Partido Liberal

→ Libre postulación

9.21 Union membership

PA_L3_AFILIACION_SINDICAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Membership of a union, guild or workers' association, union dues, collective agreement.

HOW IT IS FOUND

COLUMN NAME

PA_L3_AFILIACION_SINDICAL - Path

95%

sindicato[a-z]* sindical sindicalizado afiliaci[oó]n?sindical cuota?sindical
delegado?sindical seccional?sindical gremio gremial asociacion?(de_)?
(docentes|educadores|profesores|empleados) union?member(ship)? trade?union convencion?
colectiva

KNOWN VALUES

PA_L3_AFILIACION_SINDICAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-sindicatos.txt

15

60%

SUNTRACS, CONATO, CONUSI, CONATLAB, ASOPROF, A EVE

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Flags keep their shape; any other value becomes Sin información .

PA_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_SUPRIMIR .

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

SUNTRACS

→

Sin información

Economic and credit data

The law does not call them sensitive, but it requires prior consent for economic, financial, banking and commercial data (art. 8, 3), protects data subjects from automated decisions about their credit (art. 19), and credit references have a law of their own, Law 24 of 2002. They go beyond the bank account: an APC credit reference, a pension amount, a MIDES program or a poverty status say as much about a person as their salary. Categories are replaced within their own notation; amounts are mapped into a plausible range unrelated to the real value.

10.1 Credit reference

PA_FIN_REFERENCIA_CREDITO

COLUMN NAME

REGEX DECOMPOSE

WHAT IT IS

APC scores and references, days past due, delinquent portfolio, collection lawsuits, written-off accounts.

HOW IT IS FOUND

COLUMN NAME

PA_FIN_REFERENCIA_CREDITO – Path

90%

```
apc referencia_(de_)?credito referencias_crediticias historial_(de_)?credito
historial_crediticio score(_?(apc|crediticio|credito|riesgo))? puntaje_
(crediticio|apc|riesgo|credito) calificacion_(crediticia|apc|riesgo) morosidad moroso
morosa dias_(de_)?(atraso|mora) cartera_(vencida|morosa) cobro_judicial
castigad[oa] estatus_(credito|crediticio|cartera) estado_(de_)?(cuenta_)?
(credito|crediticio|cartera) credit_score delinquen[a-z]*
```

HOW IT IS MASKED

A score gets other digits, a flag stays a flag and any status is replaced by another credit status.

PA_APC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_REFERENCIA_CREDITO .

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

PA_REFERENCIA_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `pa-referencia-credito.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Morosidad de 90 días

→

Morosidad de 60 días

745

→

471

10.2 Debt amount

PA_FIN_MONTO_DEUDA

COLUMN NAME

DATA TYPE

NUMERIC MAPPING

WHAT IT IS

Balances of loans, cards and mortgages, past-due amounts, credit limits, installments.

HOW IT IS FOUND

COLUMN NAME

PA_FIN_MONTO_DEUDA - Path

85%

monto_?(deuda|adeudado|vencido|moroso|credito|prestamo|financiado|hipoteca) deuda(_?
(total|vigente|vencida))? adeudo saldo_
(deudor|insoluto|vencido|moroso|credito|prestamo|tarjeta|hipoteca) limite_?(de_)?credito
linea_?(de_)?credito pago_?minimo letra_?(mensual|prestamo|hipoteca) debt(_?amount)?
outstanding_?balance

DATA TYPE

PA_FIN_MONTO_DEUDA - Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to an amount between B/.100 and B/.500,000.

PA_MONTO_DEUDA

NUMERIC MAPPING

Maps the number to another between 100 and 500,000.

EXAMPLES

VALUE

MASKED

8500

→

5134

10.3 Salary and income

PA_FIN_INGRESO_MENSUAL

COLUMN NAME

DATA TYPE

NUMERIC MAPPING

WHAT IT IS

Salary, thirteenth-month pay, household income.

HOW IT IS FOUND

COLUMN NAME

PA_FIN_INGRESO_MENSUAL - Path

85%

salario[a-z]* sueldo[a-z]* remuneraci[oó]n[a-z]* ingreso_
(mensual|neto|bruto|familiar|del_?hogar|hogar|per_
capita|total|promedio|declarado|laboral|anual) ingresos(_?[a-z]+)? decimo_?tercer_?mes
xiii_?mes total_?devengado neto_?a_?pagar planilla_?(neta|bruta) gasto_
(mensual|familiar) capacidad_?(de_)?pago salary wage income gross_?pay net_?pay

DATA TYPE PA_FIN_INGRESO_MENSUAL – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to an amount between B/.350 and B/.25,000.

PA_INGRESO_MENSUAL **NUMERIC MAPPING**

Maps the number to another between 350 and 25,000.

EXAMPLES

VALUE		MASKED
1850	→	1958

10.4 Assets

PA_FIN_PATRIMONIO

COLUMN NAME **DATA TYPE** **NUMERIC MAPPING**

WHAT IT IS

Cadastral or market value of real estate and vehicles, savings, term deposits, net worth.

HOW IT IS FOUND

COLUMN NAME PA_FIN_PATRIMONIO – Path

85% patrimonio(_?(neto|total))? valor_
(catastral|comercial|avaluo|inmueble|propiedad|finca|vivienda|vehiculo|de_la_garantia)
avaluo bienes_(inmuebles|muebles) valor_bienes activos_(personales|totales) saldo_
(inversion|ahorro|ahorros|cuenta|plazo_fijo) monto_(inversion|ahorro|plazo_fijo) net_
worth assets

DATA TYPE PA_FIN_PATRIMONIO – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to an amount between B/.10,000 and B/.3,000,000.

PA_PATRIMONIO **NUMERIC MAPPING**

Maps the number to another between 10,000 and 3,000,000.

EXAMPLES

VALUE		MASKED
185000	→	477515

10.5 Pension and individual account

PA_FIN_PENSION

COLUMN NAME **DATA TYPE** **NUMERIC MAPPING**

WHAT IT IS

CSS retirement or pension amount, balance of the individual account, of SIACAP or of the severance fund.

HOW IT IS FOUND

COLUMN NAME PA_FIN_PENSION – Path

90% monto_?(pension|jubilacion) pension_?(mensual|monto|vejez|invalides|sobreviviente)
jubilacion_?(mensual|monto) saldo_?(cuenta_?individual|siacap|retiro|fondo_
complementario) cuenta_?individual siacap fondo_?(de_?)?cesantia prima_?de_?antiguedad
retirement_?balance

DATA TYPE PA_FIN_PENSION – Type

Accepts number; any other type rules the domain out.

HOW IT IS MASKED

Mapped to an amount between B/.120 and B/.6,000.

PA_MONTO_PENSION **NUMERIC MAPPING**

Maps the number to another between 120 and 6,000.

EXAMPLES

VALUE	MASKED
680	→ 227

10.6 Social security coverage

PA_FIN_SEGURO_SOCIAL

COLUMN NAME **KNOWN VALUES** **REGEX DECOMPOSE**

WHAT IT IS

A person's standing with the Social Security Fund, in the census categories: direct insured, beneficiary, retired or pensioner, no social security. It reveals formal or informal work.

HOW IT IS FOUND

COLUMN NAME PA_FIN_SEGURO_SOCIAL – Path

90% tipo_?(de_?)?(asegurado|seguro_?social|cobertura) condicion_?(de_?)?asegurado cobertura_
(css|salud|seguro_?social) afiliacion_?(a_?la_?)?(css|caja_?de_?seguro_?social|seguridad_
social) asegurado_?css es_?asegurado seguro_?(medico|de_?salud|privado) health_
insurance

KNOWN VALUES PA_FIN_SEGURO_SOCIAL – List

LIST	VALUES	STRENGTH	FIRST VALUES
pa-detectar-seguro-social.txt	20	80%	Asegurado directo, Beneficiario, Jubilado o pensionado por vejez, Pensionado por enfermedad o accidente, Jubilado o pensionado de otro país, Jubilado o pensionado de otro país

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the category is replaced by another.

PA_SEGURO_SOCIAL **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_SEGURO_SOCIAL_LISTA .

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_SEGURO_SOCIAL_LISTA` **SECURE LOOKUP**

Replaces the value with one of the 6 values in `pa-seguro-social.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Beneficiario

→ Jubilado o pensionado de otro país

10.7 Social program

PA_FIN_PROGRAMA_SOCIAL

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Beneficiaries of transfer programs: MIDES's 120 a los 65, Red de Oportunidades, Ángel Guardián and Bono Alimenticio Nutricional, IFARHU's PASE-U and scholarships. Being a beneficiary reveals poverty, old age or disability.

HOW IT IS FOUND

COLUMN NAME

PA_FIN_PROGRAMA_SOCIAL - Path

90%

```
programa_?(social|de_?transferencia|de_?apoyo|gobierno|mides) programas_?sociales
beneficiario_?(de_?)?(programa|bono|beca|subsidio|mides) bono_?
(solidario|alimenticio|nutricional|digital) vale_?digital subsidio_?
(estatal|social|senadis) transferencia_?monetaria red_?(de_?)?oportunidades angel_?
guardian 120_?a_?los_?65 pase_?u beca_?universal ifarhu senapan padron_?(de_?)?
beneficiarios social_?program
```

KNOWN VALUES

PA_FIN_PROGRAMA_SOCIAL - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-
programas.txt

23

80%

120 a los 65, Red de Oportunidades, Ángel Guardián, Angel Guardian, Bono Alimenticio Nutricional, PASE-U

A value that matches none counts 40% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the program is replaced by another.

`PA_PROGRAMA_SOCIAL` **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_PROGRAMA_SOCIAL_LISTA` .

`PA_BANDERA` **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

`PA_BANDERA_SI_NO` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt`, chosen by a keyed hash.

`PA_BANDERA_TRUE_FALSE` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt`, chosen by a keyed hash.

`PA_BANDERA_S_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt`, chosen by a keyed hash.

`PA_BANDERA_Y_N` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt`, chosen by a keyed hash.

`PA_BANDERA_0_1` **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt`, chosen by a keyed hash.

`PA_CM_ALFANUM` **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

`PA_PROGRAMA_SOCIAL_LISTA` **SECURE LOOKUP**

Replaces the value with one of the 9 values in `pa-programas-sociales.txt`, chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Vale Digital

→ Auxilio económico IFARHU

10.8 Socioeconomic level

`PA_FIN_NIVEL_SOCIOECONOMICO`

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Poverty status, income quintile or decile, multidimensional poverty index.

HOW IT IS FOUND

COLUMN NAME

`PA_FIN_NIVEL_SOCIOECONOMICO` - Path

90%

`nivel_?socioeconomico nivel_?socio_?economico estrato(_?socioeconomico)? clase_?social
decil(_?(de_?)?ingresos)? quintil(_?(de_?)?ingresos)? pobreza(_?
(extrema|general|multidimensional))? indice_?(de_?)?pobreza ipm vulnerabilidad(_?
(social|economica))? carencias? socioeconomic_?level ficha_?socioeconomica`

KNOWN VALUES

`PA_FIN_NIVEL_SOCIOECONOMICO` - List

LIST

VALUES

STRENGTH

FIRST VALUES

`pa-detectar-
pobreza.txt`

10

40%

pobreza extrema, pobreza general, pobre, no pobre, indigente,
vulnerable

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

A quintile or decile becomes another number of the same range, with its prefix; a status, another from the list.

`PA_NIVEL_SOCIOECONOMICO` **REGEX DECOMPOSE**

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_POBREZA`.

`PA_QUINTIL` **SECURE LOOKUP**

Replaces the value with one of the 5 values in `pa-quintil.txt`, chosen by a keyed hash.

`PA_DECIL_ALTO` **SECURE LOOKUP**

Replaces the value with one of the 5 values in `pa-decil-alto.txt`, chosen by a keyed hash.

PA_DECIL **SECURE LOOKUP**

Replaces the value with one of the 10 values in `pa-decil.txt` , chosen by a keyed hash.

PA_POBREZA **SECURE LOOKUP**

Replaces the value with one of the 4 values in `pa-pobreza.txt` , chosen by a keyed hash.

EXAMPLES

VALUE		MASKED
Pobreza extrema	→	Pobreza general
Quintil 2	→	Quintil 4

10.9 Housing tenure

PA_FIN_VIVIENDA

COLUMN NAME

KNOWN VALUES

REGEX DECOMPOSE

WHAT IT IS

Whether the home is owned, mortgaged, rented, lent or an informal settlement, and its conditions.

HOW IT IS FOUND

COLUMN NAME

PA_FIN_VIVIENDA - Path

95%

`tenencia_(de_)?(la_)?vivienda` `tipo_(de_)?vivienda` `situacion_(de_la_)?vivienda`
`vivienda_(propia|alquilada|prestada)` `condicion_(de_)?(la_)?vivienda` `material_(de_)?`
`(la_)?vivienda` `piso_de_tierra` `hacinamiento` `precarista` `invasion` `housing_?`
`(type|tenure|status)`

KNOWN VALUES

PA_FIN_VIVIENDA - List

LIST

VALUES

STRENGTH

FIRST VALUES

pa-detectar-vivienda.txt	17	60%	Propia, Propia hipotecada, Alquilada, Cedida o prestada, Condena, Otra situación
--------------------------	----	-----	--

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Flags and codes keep their shape; the tenure is replaced by another from the census list.

PA_VIVIENDA **REGEX DECOMPOSE**

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_VIVIENDA_LISTA .

PA_BANDERA **REGEX DECOMPOSE**

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

PA_BANDERA_S_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

PA_BANDERA_Y_N **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

PA_BANDERA_0_1 **SECURE LOOKUP**

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

PA_CM_ALFANUM **CHARACTER MAPPING**

Maps each character to another of its group (7 groups), keeping length and format.

PA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in `pa-vivienda.txt` , chosen by a keyed hash.

EXAMPLES

VALUE

MASKED

Invasión

→

Propia hipotecada

Criminal records

Criminal convictions have their own rules in the law (art. 30) and its regulation (Decree 285, art. 7), and are among the data most likely to give rise to discrimination. They get the same care as sensitive data.

11.1 Criminal record

PA_PENAL_ANTECEDENTES

COLUMN NAME REGEX DECOMPOSE

WHAT IT IS

Police record (*récord policivo*), criminal record, sentences, offences, detentions and precautionary measures.

HOW IT IS FOUND

COLUMN NAME PA_PENAL_ANTECEDENTES – Path

95%

antecedentes(_?(penales|judiciales|policivos|policiales|criminales))? *(except when followed by list E8)*
record_(policivo|policial|penal|criminal) certificado_(de_)?antecedentes sentencia(_?
condenatoria)? sentenciad[oa] condena(do|da)? delitos? tipo_(de_)?delito reincidente
reincidencia imputad[oa] privado_de_libertad detencion_provisional medida_cautelar
centro_penitenciario pena reclusion libertad_condicional criminal_record
conviction offen[cs]e detenid[oa] violencia_domestica

HOW IT IS MASKED

A flag stays a flag; any other value becomes a generic status: with a record, without a record, or no information.

PA_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_ANTECEDENTES_LISTA .

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-si-no.txt , chosen by a keyed hash.

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-true-false.txt , chosen by a keyed hash.

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-s-n.txt , chosen by a keyed hash.

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-y-n.txt , chosen by a keyed hash.

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in pa-bandera-0-1.txt , chosen by a keyed hash.

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

PA_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 3 values in pa-antecedentes.txt , chosen by a keyed hash.

EXAMPLES

VALUE	MASKED
Condenado por hurto	→ Sin antecedentes penales

11.2 Case number

PA_PENAL_CAUSA

COLUMN NAME	VALUE FORMAT	CHARACTER MAPPING
-------------	--------------	-------------------

WHAT IT IS

Case file (*carpetilla*), investigation folder, case, court file or complaint numbers.

HOW IT IS FOUND

COLUMN NAME	PA_PENAL_CAUSA – Path		
90%	carpetilla n(o um ro)?carpetilla carpeta?(de_)?investigacion n(o um ro)?carpeta causa?penal n(o um ro umero)?(de_)?causa expediente?(judicial penal) n(o um ro)? expediente?(judicial penal) n(o um ro)?(de_)?denuncia folio?denuncia n(o um ro)? proceso case?number		
VALUE FORMAT	PA_PENAL_CAUSA – Regex		
PATTERN	STRENGTH	NOTES	
\d{12,15}		40%	
\d{1,6}–\d{4}		50%	

A value that matches none counts 30% against the domain.

HOW IT IS MASKED

Character by character, keeping length and format.

PA_CM_ALFANUM	CHARACTER MAPPING
---------------	-------------------

Maps each character to another of its group (7 groups), keeping length and format.

EXAMPLES

VALUE	MASKED
202400012345	→ 917747298420

The RUC check digit, computed by the set

A natural person's RUC check digit is two digits that the DGI computes over a 21-position table: the cédula fields right-aligned in fixed widths — asiento 5, tomo 3, a two-digit code for the letters (AV 15, PI 79, E 50, N 40, PE 75), province 2 — and a constant 5. Weights grow from 2 at the right, modulo 11, and a remainder of 0 or 1 gives 0; the second digit repeats the computation with the first one appended. A field longer than its width pushes the fields to its left, and a one-digit province with AV or PI is padded with two zeros.

No framework of the plugin does this: Check Digit can add weighted digits, but not pad fields or weigh letters. The set does it with tables:

- The cédula is masked with the same algorithm as the cédula domain. Letters and separators are swapped for one-character markers, and the old DV is replaced by an initial state.
- The state — weighted sum, digit sum and position — travels through the value from right to left, one character per step. Each step is a Data Cleansing that turns *character + state* into *new state + character*.
- At the left end, a table turns the state into the two digits of the DV, and a second pass carries them back to where the old DV was.
- The markers become letters and separators again.

The tables are Data Cleansing files generated with the set, and the steps are Regex Decompose algorithms joined in String Algorithm Chains of at most eight — the engine's limit. Every masked RUC has a valid DV and the mapping stays one-to-one. The price is a long algorithm tree under the domain and a step table of more than 30,000 lines.

Where people live: districts and corregimientos

With the data of the 2023 XII Population Census (INEC), the set knows the population, district, province and location of each of the 699 corregimientos.

- A **corregimiento** under 10,000 inhabitants becomes the nearest corregimiento of its province or comarca with at least 10,000. Darién, Los Santos and the Emberá-Wounaan and Ngäbe-Buglé comarcas have none: theirs become the nearest one in the country.
- A **district** under 20,000 inhabitants becomes the nearest district of its province or comarca with at least 20,000, or of the country in the Emberá-Wounaan comarca.
- Distance is measured between centroids, a district's weighted by the population of its corregimientos.
- Sixty corregimiento names repeat — there are five San José —: a name is generalized only when every place bearing it is small. INEC codes have no such ambiguity.

The result is always a real place, near the original, and every group of people who end up sharing a name holds at least 10,000 inhabitants. Names that are also given names or surnames are left out of the detection list, so that a surname column is not taken for corregimientos.

Limits

- **Binary columns** — photos of the cédula, fingerprint images, clinical PDFs in BLOBs — cannot be masked by these algorithms. Inventory them and report them as residual risk.
- **A DV in its own column** is not recomputed: it is masked, and no longer matches the masked RUC. Where the application validates a RUC and DV stored separately, validation fails.
- **The cédula letters stay.** `PI` still marks a registration as an indigenous Panamanian, and `E`, `N` and `PE` the status of foreigner, naturalized or born abroad. Changing them would break the format or make different cédulas collide. If that is sensitive in your case, handle it together with the ethnic origin or nationality column.
- **Cédulas with a zero-padded province** (`08-442-445`) or without hyphens are masked character by character, without recomputing the DV. An `NT` RUC is also masked without recomputing.
- **A legal entity's RUC stays.** A company with a single owner is still a legal entity under the law.
- **Data Cleansing leaves unknown values as they are:** a corregimiento spelled in a way the list does not have passes unchanged.
- **Free text** is redacted token by token: a phone written with spaces is not found, and a given name that is also a common word (`Luz` , `Rosa`) is redacted wherever it appears.
- **Minors** — a birth date moves at most 60 days, so few people cross their 18th birthday, but some do. Where that matters, split the rule by age group with Multi-Column Condition.
- **Addresses** are replaced as a whole line. Street, *barriada*, *corregimiento* and district in separate columns are masked independently and no longer form a coherent address.
- **Plates and VINs** keep their format, but not the VIN check digit.

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Verification

Before publication the set was checked outside a Masking Engine:

- The computation of the RUC check digit was first checked against the examples of the published algorithm.
- Every classifier configuration was reviewed against what a Masking Engine accepts.
- Profiling was simulated on 138 columns with the whole set at its assignment threshold — well named, badly named, and 24 columns that are not personal data — and again with the classifiers of the essential pack alone.
- The algorithms, run with the Delphix algorithm plugin, masked 255 sample values and 210 placeholders such as `N/D` : RUCs with a DV in the four separators and the five letter types, checking each DV; 150 cédulas and 150 RUCs for collisions; and 20 people whose cédula and RUC must still agree.

Every column was assigned as expected and every value masked as expected. Before using the set on real data, run a profiling job and a masking job on a representative copy.

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Sources

- Law 81 of 26 March 2019, on Personal Data Protection — Gaceta Oficial 28743-A, 29 March 2019 (National Assembly, Legispan).

- Executive Decree 285 of 28 May 2021, regulating Law 81 of 2019.
- National Authority for Transparency and Access to Information (ANTAI), Personal Data Protection Directorate.
- Algorithm for the RUC check digit — Dirección General de Ingresos and the National Authority for Government Innovation.
- Format of the cédula and its letters — Electoral Tribunal, National Directorate of Identity Cards. Party membership, July and August 2026 — Electoral Tribunal.
- Population, code and location of the 699 corregimientos; indigenous and Afro-descendant groups; marital status; social security; schooling — XII National Population and VIII Housing Census 2023 (INEC), in the geographic service by corregimiento published with Esri Panamá.
- Most frequent surnames and given names — Civil Registry of the Electoral Tribunal.
- Certified disability types — National Secretariat for Disability (SENADIS). Transfer programs, 2026 — Ministry of Social Development (MIDES) and IFARHU (PASE-U).
- Credit references — Law 24 of 2002 and APC. Religion — 2015 national survey. Plates — Land Transit and Transport Authority (2013 format). Phone numbering — ASEP, National Numbering Plan. Postal codes — Correos y Telégrafos.

APPENDIX A

Algorithms

Every algorithm of the set in alphabetical order, with its configuration. The one a domain names is the top of its tree; the others are building blocks.

PA_ANIO REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_L2_ANIO_NACIMIENTO

PATTERNS, TRIED IN ORDER

(\d{3})(\d)	keep · PA_CM_ALFANUM
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_ANTECEDENTES REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_ANTECEDENTES_LISTA .

Used by: PA_PENAL_ANTECEDENTES

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_ANTECEDENTES_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_ANTECEDENTES_LISTA SECURE LOOKUP

Replaces the value with one of the 3 values in pa-antecedentes.txt , chosen by a keyed hash.

Used by: PA_ANTECEDENTES

CONFIGURATION

lookupFile	pa-antecedentes.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_APC REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_REFERENCIA_CREDITO .

Used by: PA_FIN_REFERENCIA_CREDITO

PATTERNS, TRIED IN ORDER

(\d{2,3})	PA_CM_ALFANUM
(?i)(s[ií] no true false [sny01])	PA_BANDERA
otherwise	PA_REFERENCIA_CREDITO

CONFIGURATION

trimInput	yes
requireMask	no

PA_APELLIDO NAME

Replaces the name with one of the 200 names in `pa-apellidos.txt`.

Used by: `PA_APELLIDOS`, `PA_NOMBRE_COMPLETO`

CONFIGURATION

lookupFile	pa-apellidos.txt
maskedValueCase	PRESERVE_INPUT
filterAccent	yes
inputCaseSensitive	no

PA_APELLIDOS REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM`.

Used by: `PA_L1_APELLIDO`, `PA_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

(\S+)	PA_APELLIDO
(\S+)\s+(\S+)	PA_APELLIDO · PA_APELLIDO
(\S+)(\s+)([Dd][Ee] [Vv][Dd][Aa]\.?\s+[Dd][Ee])(\s+)(\S+)	PA_APELLIDO · keep · keep · keep · PA_APELLIDO
(\S+)\s+(\S+)\s+(\S+)	PA_APELLIDO · PA_APELLIDO · PA_APELLIDO
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PA_BANDERA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: `keep`.

Used by: `PA_CATEGORIA_SUPRIMIDA`, `PA_PUEBLO`, `PA LENGUA_INDIGENA`, `PA_DIAGNOSTICO`, `PA_MEDICAMENTO`, `PA_DISCAPACIDAD`, `PA_ORIENTACION_SEXUAL`, `PA_IDENTIDAD_GENERO`, `PA_RELIGION`, `PA_OPINION_POLITICA`, `PA_PARTIDO_POLITICO`, `PA_APC`, `PA_SEGURO_SOCIAL`, `PA_PROGRAMA_SOCIAL`, `PA_VIVIENDA`, `PA_ANTECEDENTES`

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no)	PA_BANDERA_SI_NO
(?i)(true false)	PA_BANDERA_TRUE_FALSE
(?i)([sn])	PA_BANDERA_S_N
(?i)(y)	PA_BANDERA_Y_N
([01])	PA_BANDERA_0_1
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_BANDERA_0_1 SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-0-1.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

CONFIGURATION

lookupFile	<code>pa-bandera-0-1.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_BANDERA_S_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-s-n.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

CONFIGURATION

lookupFile	<code>pa-bandera-s-n.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_BANDERA_SI_NO SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-si-no.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

CONFIGURATION

lookupFile	<code>pa-bandera-si-no.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_BANDERA_TRUE_FALSE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-true-false.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

CONFIGURATION

lookupFile	<code>pa-bandera-true-false.txt</code>
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT

inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_BANDERA_Y_N SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-bandera-y-n.txt` , chosen by a keyed hash.

Used by: `PA_BANDERA`

CONFIGURATION

lookupFile	<code>pa-bandera-y-n.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_BARRIADA SECURE LOOKUP

Replaces the value with one of the 75 values in `pa-barriadas.txt` , chosen by a keyed hash.

Used by: `PA_L2_BARRIADA`

CONFIGURATION

lookupFile	<code>pa-barriadas.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_LOOKUP_FILE</code>
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_CANTIDAD_ACOTADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L2_DEPENDIENTES`

PATTERNS, TRIED IN ORDER

<code>([0-5])</code>	keep
<code>(\d+)</code>	replace with <code>5</code>
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_CATEGORIA_SUPRIMIDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_SUPRIMIR` .

Used by: `PA_L3_SALUD_SEXUAL_REPRODUCTIVA` , `PA_L3_SALUD_MENTAL_ADICCIONES` , `PA_L3_VIDA_SEXUAL` , `PA_L3_CREENCIA_FILOSOFICA_MORAL` , `PA_L3_AFILIACION_SINDICAL`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false [sny01])</code>	<code>PA_BANDERA</code>
<code>(\d{1,6})</code>	<code>PA_CM_ALFANUM</code>

otherwise	PA_SUPRIMIR
CONFIGURATION	
trimInput	yes
requireMask	no

PA_CEDULA REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L1_CEDULA , PA_RUC

PATTERNS, TRIED IN ORDER

([1-9] 1[0-3]) ([\\s\\-]) (\\d{1,4}) ([\\s\\-]) (\\d{1,6})	PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM
([1-9] 1[0-3]) (AV PI) ([\\s\\-]) (\\d{1,4}) ([\\s\\-]) (\\d{1,6})	PA_PROVINCIA_CEDULA · keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM
(E) ([\\s\\-]) ([1-9] 1[0-3]) ([\\s\\-]) (\\d{1,6})	keep · keep · PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM
(N PE) ([\\s\\-]) (\\d{1,4}) ([\\s\\-]) (\\d{1,6})	keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM
([1-9] 1[0-3]) ([\\s\\-] NT [\\s\\-]) (\\d{1,4}) ([\\s\\-]) (\\d{1,6})	PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM
otherwise	PA_CM_ALFANUM

CONFIGURATION	
trimInput	yes
requireMask	no

PA_CIE10 SECURE LOOKUP

Replaces the value with one of the 37 values in pa-cie10.txt , chosen by a keyed hash.

Used by: PA_DIAGNOSTICO

CONFIGURATION	
lookupFile	pa-cie10.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_CIE10_DECIMAL SECURE LOOKUP

Replaces the value with one of the 33 values in pa-cie10-decimal.txt , chosen by a keyed hash.

Used by: PA_DIAGNOSTICO

CONFIGURATION	
lookupFile	pa-cie10-decimal.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_CM_ALFANUM CHARACTER MAPPING

Maps each character to another of its group (7 groups), keeping length and format.

Used by: PA_L1_PASAPORTE , PA_L1_DOCUMENTO_MIGRATORIO , PA_L1_SEGURO_SOCIAL , PA_L1_IDONEIDAD , PA_L1_DIRECCION_COMPLEMENTO , PA_L1_CUENTA_BANCARIA , PA_L1_VIN , PA_L1_FINCA , PA_L1_NUMERO_CONTRATO , PA_L1_USUARIO_RRSS , PA_L3_SALUD_IDENTIFICADOR , PA_L3_SALUD_RESULTADO_EXAMEN , PA_PENAL_CAUSA , PA_CATEGORIA_SUPRIMIDA , PA_CEDULA , PA_RUC_DV_PREPARAR , PA_RUC , PA_DV , PA_NOMBRES , PA_APELLIDOS , PA_NOMBRE_COMPLETO , PA_EMAIL , PA_TELEFONO , PA_PLACA , PA_COORDENADA , PA_ANIO , PA_EDAD , PA_CODIGO_POSTAL , PA_PUEBLO , PA LENGUA_INDIGENA , PA_MEDICAMENTO , PA_DISCAPACIDAD , PA_ORIENTACION_SEXUAL , PA_IDENTIDAD_GENERO , PA_RELIGION , PA_OPINION_POLITICA , PA_PARTIDO_POLITICO , PA_APC , PA_SEGURO_SOCIAL , PA_PROGRAMA_SOCIAL , PA_VIVIENDA , PA_ANTECEDENTES

CONFIGURATION

characterGroups	0123456789 AEIOU BCDFGHJKLMNPQRSTUVWXYZ aeiou bcdfghjklmnpqrstvwxyz ÁÉÍÓÚÚ áéíóúú
caseSensitive	yes
minMaskedPositions	0

PA_CM_HEX CHARACTER MAPPING

Maps each character to another of its group (5 groups), keeping length and format.

Used by: PA_L1_DISPOSITIVO , PA_IP

CONFIGURATION

characterGroups	0123456789 abcdef ABCDEF ghijklmnopqrstuvwxyz GHIJKLMNOPQRSTUVWXYZ
caseSensitive	yes
minMaskedPositions	0

PA_CODIGO_CORREGIMIENTO DATA CLEANSING

Replaces each value found in pa-codigo-corregimiento-generalizado.txt (1,331 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA_CODIGO_LUGAR

CONFIGURATION

lookupFile	pa-codigo-corregimiento-generalizado.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_CODIGO_DISTRITO DATA CLEANSING

Replaces each value found in pa-codigo-distrito-generalizado.txt (148 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA_CODIGO_LUGAR

CONFIGURATION

lookupFile	pa-codigo-distrito-generalizado.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_CODIGO_LUGAR REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA_L2_CODIGO_LUGAR

PATTERNS, TRIED IN ORDER

(\d{5,6})	PA_CODIGO_CORREGIMIENTO
(\d{3,4})	PA_CODIGO_DISTrito
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_CODIGO_POSTAL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L2_CODIGO_POSTAL

PATTERNS, TRIED IN ORDER

(\d{2})(\d{2,3})	keep · each character → 0
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PA_COORDENADA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA_L1_GEOLOCALIZACION

PATTERNS, TRIED IN ORDER

(-?\d{1,3}\.\d)(\d+)\s*,\s*(-?\d{1,3}\.\d)(\d+)	keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM
(-?\d{1,3}\.\d)(\d+)	keep · PA_CM_ALFANUM
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_CORREGIMIENTO DATA CLEANSING

Replaces each value found in pa-corregimiento-generalizado.txt (1,760 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA_L2_CORREGIMIENTO

CONFIGURATION

lookupFile	pa-corregimiento-generalizado.txt
delimiter	
caseSensitive	yes
trimWhitespace	yes

PA_DECIL SECURE LOOKUP

Replaces the value with one of the 10 values in pa-decil.txt , chosen by a keyed hash.

Used by: PA_NIVEL_SOCIOECONOMICO

CONFIGURATION

lookupFile	pa-decil.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_DECIL_ALTO SECURE LOOKUP

Replaces the value with one of the 5 values in pa-decil-alto.txt , chosen by a keyed hash.

Used by: PA_NIVEL_SOCIOECONOMICO

CONFIGURATION

lookupFile	pa-decil-alto.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_DIAGNOSTICO REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches:

PA_DIAGNOSTICO_TEXTO .

Used by: PA_L3_SALUD_DIAGNOSTICO

PATTERNS, TRIED IN ORDER

([A-TV-Za-tv-z]\d{2}\.\d{1,2})	PA_CIE10_DECIMAL
([A-TV-Za-tv-z]\d{2,3})	PA_CIE10
(?i)(s[ií] no true false [sny01])	PA_BANDERA
otherwise	PA_DIAGNOSTICO_TEXTO

CONFIGURATION

trimInput	yes
requireMask	no

PA_DIAGNOSTICO_TEXTO SECURE LOOKUP

Replaces the value with one of the 30 values in pa-diagnostics.txt , chosen by a keyed hash.

Used by: PA_DIAGNOSTICO

CONFIGURATION

lookupFile	pa-diagnostics.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_DIRECCION SECURE LOOKUP

Replaces the value with one of the 3,000 values in `pa-direcciones.txt` , chosen by a keyed hash.

Used by: `PA_L1_DIRECCION`

CONFIGURATION

lookupFile	pa-direcciones.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_DISCAPACIDAD REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_DISCAPACIDAD_LISTA` .

Used by: `PA_L3_SALUD_DISCAPACIDAD`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false [sny01])</code>	<code>PA_BANDERA</code>
<code>(\d{1,6})</code>	<code>PA_CM_ALFANUM</code>
otherwise	<code>PA_DISCAPACIDAD_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

PA_DISCAPACIDAD_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in `pa-discapacidad.txt` , chosen by a keyed hash.

Used by: `PA_DISCAPACIDAD`

CONFIGURATION

lookupFile	pa-discapacidad.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_DISTRITO DATA CLEANSING

Replaces each value found in `pa-distrito-generalizado.txt` (234 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PA_L2_DISTRITO`

CONFIGURATION

lookupFile	pa-distrito-generalizado.txt
delimiter	
caseSensitive	yes
trimWhitespace	yes

PA_DV REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L1_RUC_DV

PATTERNS, TRIED IN ORDER

(\d{1,2})	PA_CM_ALFANUM
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PA_EDAD REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA_L2_EDAD

PATTERNS, TRIED IN ORDER

(\d{3,})	replace with 90
([1-9])(\d)(\.\d+)	keep · PA_CM_ALFANUM · keep
([1-9])(\d)	keep · PA_CM_ALFANUM
(\d)	PA_CM_ALFANUM
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_EMAIL REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L1_EMAIL

PATTERNS, TRIED IN ORDER

([^\s@]+)@([^\s@]+\.\.[A-Za-z]{2,24})	PA_CM_ALFANUM · PA_CM_ALFANUM · replace with test
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PA_EMPLEADOR SECURE LOOKUP

Replaces the value with one of the 25 values in pa-empleadores.txt , chosen by a keyed hash.

Used by: PA_L2_EMPLEADOR

CONFIGURATION

lookupFile	pa-empleadores.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes

trimWhitespaceInLookupFile	yes
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PA_ESCOLARIDAD SECURE LOOKUP

Replaces the value with one of the 13 values in `pa-escolaridad.txt`, chosen by a keyed hash.

Used by: `PA_L2_ESCOLARIDAD`

CONFIGURATION

lookupFile	<code>pa-escolaridad.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

PA_ESCUELA SECURE LOOKUP

Replaces the value with one of the 15 values in `pa-escuelas.txt`, chosen by a keyed hash.

Used by: `PA_L2_ESCUELA`

CONFIGURATION

lookupFile	<code>pa-escuelas.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

PA_ESTADO_CIVIL SECURE LOOKUP

Replaces the value with one of the 7 values in `pa-estado-civil.txt`, chosen by a keyed hash.

Used by: `PA_L2_ESTADO_CIVIL`

CONFIGURATION

lookupFile	<code>pa-estado-civil.txt</code>
hashMethod	<code>SHA256</code>
maskedValueCase	<code>PRESERVE_INPUT</code>
inputCaseSensitive	<code>no</code>
trimWhitespaceFromInput	<code>yes</code>
trimWhitespaceInLookupFile	<code>yes</code>

PA_FECHA_EVENTO DATE SHIFT

Moves the date by -30 to 30 days.

Used by: `PA_L2_FECHA_EVENTO`

CONFIGURATION

minRange	<code>-30</code>
maxRange	<code>30</code>
unit	<code>DAYS</code>
roll	<code>no</code>

PA_FECHA_NACIMIENTO DATE SHIFT

Moves the date by -60 to 60 days.

Used by: PA_L2_FECHA_NACIMIENTO

CONFIGURATION

minRange	-60
maxRange	60
unit	DAYS
roll	no

PA_GRUPO_SANGUINEO SECURE LOOKUP

Replaces the value with one of the 8 values in pa-grupo-sanguineo.txt , chosen by a keyed hash.

Used by: PA_L3_SALUD_GRUPO_SANGUINEO

CONFIGURATION

lookupFile	pa-grupo-sanguineo.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_IDENTIDAD_GENERO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_IDENTIDAD_GENERO_LISTA .

Used by: PA_L3_IDENTIDAD_GENERO

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_IDENTIDAD_GENERO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_IDENTIDAD_GENERO_LISTA SECURE LOOKUP

Replaces the value with one of the 8 values in pa-identidad-genero.txt , chosen by a keyed hash.

Used by: PA_IDENTIDAD_GENERO

CONFIGURATION

lookupFile	pa-identidad-genero.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_INGRESO_MENSUAL NUMERIC MAPPING

Maps the number to another between 350 and 25,000.

Used by: PA_FIN_INGRESO_MENSUAL

CONFIGURATION

minValue	350
maxValue	25000

PA_IP REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: PA_CM_HEX .

Used by: PA_L1_IP

PATTERNS, TRIED IN ORDER

(\d{1,3})\.\(\d{1,3})\.\(\d{1,3})\.\(\d{1,3})	PA_OCTETO · PA_OCTETO · PA_OCTETO · PA_OCTETO
otherwise	PA_CM_HEX

CONFIGURATION

trimInput	yes
requireMask	no

PA LENGUA INDIGENA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA LENGUA INDIGENA LISTA .

Used by: PA_L3 LENGUA INDIGENA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA LENGUA INDIGENA LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA LENGUA INDIGENA LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in pa-lenguas.txt , chosen by a keyed hash.

Used by: PA LENGUA INDIGENA

CONFIGURATION

lookupFile	pa-lenguas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_LUGAR_POBLADO SECURE LOOKUP

Replaces the value with one of the 52 values in pa-lugares-poblados.txt , chosen by a keyed hash.

Used by: PA_L2_LUGAR_POBLADO

CONFIGURATION

lookupFile	pa-lugares-poblados.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_MEDICAMENTO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_MEDICAMENTO_LISTA .

Used by: PA_L3_SALUD_MEDICAMENTO

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_MEDICAMENTO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_MEDICAMENTO_LISTA SECURE LOOKUP

Replaces the value with one of the 26 values in pa-medicamentos.txt , chosen by a keyed hash.

Used by: PA_MEDICAMENTO

CONFIGURATION

lookupFile	pa-medicamentos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_MONTO_DEUDA NUMERIC MAPPING

Maps the number to another between 100 and 500,000.

Used by: PA_FIN_MONTO_DEUDA

CONFIGURATION

minValue	100
maxValue	500000

PA_MONTO_PENSION NUMERIC MAPPING

Maps the number to another between 120 and 6,000.

Used by: PA_FIN_PENSION

CONFIGURATION

minValue	120
maxValue	6000

PA_NACIONALIDAD SECURE LOOKUP

Replaces the value with one of the 20 values in `pa-nacionalidades.txt` , chosen by a keyed hash.

Used by: `PA_L2_NACIONALIDAD`

CONFIGURATION

lookupFile	pa-nacionalidades.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_NIVEL_SOCIOECONOMICO REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: `PA_POBREZA` .

Used by: `PA_FIN_NIVEL_SOCIOECONOMICO`

PATTERNS, TRIED IN ORDER

<code>([1-5])</code>	<code>PA_QUINTIL</code>
<code>([6-9] 10)</code>	<code>PA_DECIL_ALTO</code>
<code>(?i)(decil\s*)(\d{1,2})</code>	<code>keep · PA_DECIL</code>
<code>(?i)(quintil\s*)([1-5])</code>	<code>keep · PA_QUINTIL</code>
otherwise	<code>PA_POBREZA</code>

CONFIGURATION

trimInput	yes
requireMask	no

PA_NOMBRE_COMPLETO REGEX DECOMPOSE

Tries 8 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

<code>([^,]+), \s*(.+)</code>	<code>PA_APELLIDOS · PA_NOMBRES</code>
<code>(\S+)</code>	<code>PA_NOMBRE_PILA</code>
<code>(\S+) \s+ (\S+)</code>	<code>PA_NOMBRE_PILA · PA_APELLIDO</code>
<code>(\S+) (\S+) (\S+) (\S+) ([Dd] [Ee] [Vv] [Dd] [Aa] \. ? \s+ [Dd] [Ee]) (\S+) (\S+)</code>	<code>PA_NOMBRE_PILA · keep · PA_APELLIDO · keep · keep · keep · PA_APELLIDO</code>
<code>(\S+) \s+ (\S+) \s+ (\S+)</code>	<code>PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>
<code>(\S+) (\S+) (\S+) (\S+) (\S+) (\S+) ([Dd] [Ee] [Vv] [Dd] [Aa] \. ? \s+ [Dd] [Ee]) (\S+) (\S+)</code>	<code>PA_NOMBRE_PILA · keep · PA_NOMBRE_PILA · keep · PA_APELLIDO · keep · keep · keep · PA_APELLIDO</code>
<code>(\S+) \s+ (\S+) \s+ (\S+) \s+ (\S+)</code>	<code>PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>
<code>(\S+) \s+ (\S+) \s+ (\S+) \s+ (\S+) \s+ (\S+)</code>	<code>PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_NOMBRE_PILA · PA_APELLIDO · PA_APELLIDO</code>
otherwise	<code>PA_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
requireMask	no

PA_NOMBRE_PILA NAME

Replaces the name with one of the 253 names in `pa-nombres.txt` .

Used by: `PA_NOMBRES` , `PA_NOMBRE_COMPLETO`

CONFIGURATION

lookupFile	pa-nombres.txt
maskedValueCase	PRESERVE_INPUT
filterAccent	yes
inputCaseSensitive	no

PA_NOMBRES REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_NOMBRE` , `PA_NOMBRE_COMPLETO`

PATTERNS, TRIED IN ORDER

<code>(\S+)</code>	<code>PA_NOMBRE_PILA</code>
<code>(\S+)\s+(\S+)</code>	<code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code>
<code>(\S+)\s+(\S+)\s+(\S+)</code>	<code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code> · <code>PA_NOMBRE_PILA</code>
otherwise	<code>PA_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
requireMask	no

PA_OCTETO NUMERIC MAPPING

Maps the number to another between 1 and 254.

Used by: `PA_IP`

CONFIGURATION

minValue	1
maxValue	254

PA_OCUPACION SECURE LOOKUP

Replaces the value with one of the 40 values in `pa-ocupaciones.txt` , chosen by a keyed hash.

Used by: `PA_L2_OCUPACION`

CONFIGURATION

lookupFile	pa-ocupaciones.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_OPINION_POLITICA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_OPINION_POLITICA_LISTA` .

Used by: `PA_L3_OPINION_POLITICA`

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_OPINION_POLITICA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_OPINION_POLITICA_LISTA SECURE LOOKUP

Replaces the value with one of the 7 values in `pa-opiniones-politicas.txt` , chosen by a keyed hash.

Used by: `PA_OPINION_POLITICA`

CONFIGURATION

lookupFile	pa-opiniones-politicas.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_ORIENTACION_SEXUAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_ORIENTACION_SEXUAL_LISTA` .

Used by: `PA_L3_ORIENTACION_SEXUAL`

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_ORIENTACION_SEXUAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_ORIENTACION_SEXUAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-orientacion-sexual.txt` , chosen by a keyed hash.

Used by: `PA_ORIENTACION_SEXUAL`

CONFIGURATION

lookupFile	pa-orientacion-sexual.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_PARTIDO_POLITICO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_PARTIDO_POLITICO_LISTA .

Used by: PA_L3_AFILIACION_POLITICA

PATTERNS, TRIED IN ORDER

(?i)(s[ii] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_PARTIDO_POLITICO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_PARTIDO_POLITICO_LISTA SECURE LOOKUP

Replaces the value with one of the 11 values in pa-partidos.txt , chosen by a keyed hash.

Used by: PA_PARTIDO_POLITICO

CONFIGURATION

lookupFile	pa-partidos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_PATRIMONIO NUMERIC MAPPING

Maps the number to another between 10,000 and 3,000,000.

Used by: PA_FIN_PATRIMONIO

CONFIGURATION

minValue	10000
maxValue	3000000

PA_PLACA REGEX DECOMPOSE

Tries 3 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L1_PLACA

PATTERNS, TRIED IN ORDER

(ADM MB MI PR BC CD CC M T B G D E)([\s-]?)	keep · keep · PA_CM_ALFANUM
(\d{3,6})	
([A-Z]{2})([\s-]?)(\d{4})	PA_CM_ALFANUM · keep · PA_CM_ALFANUM
(\d{6})	PA_CM_ALFANUM
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
requireMask	no

PA_POBREZA SECURE LOOKUP

Replaces the value with one of the 4 values in `pa-pobreza.txt` , chosen by a keyed hash.

Used by: `PA_NIVEL_SOCIOECONOMICO`

CONFIGURATION

lookupFile	pa-pobreza.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_PROGRAMA_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

`PA_PROGRAMA_SOCIAL_LISTA` .

Used by: `PA_FIN_PROGRAMA_SOCIAL`

PATTERNS, TRIED IN ORDER

<code>(?i)(s[ií] no true false [sny01])</code>	<code>PA_BANDERA</code>
<code>(\d{1,6})</code>	<code>PA_CM_ALFANUM</code>
otherwise	<code>PA_PROGRAMA_SOCIAL_LISTA</code>

CONFIGURATION

trimInput	yes
requireMask	no

PA_PROGRAMA_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 9 values in `pa-programas-sociales.txt` , chosen by a keyed hash.

Used by: `PA_PROGRAMA_SOCIAL`

CONFIGURATION

lookupFile	pa-programas-sociales.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_PROVINCIA_CEDULA NUMERIC MAPPING

Maps the number to another between 1 and 13.

Used by: `PA_CEDULA` , `PA_RUC_DV_PREPARAR` , `PA_RUC`

CONFIGURATION

minValue	1
maxValue	13

PA_PUEBLO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_PUEBLO_LISTA` .

Used by: `PA_L3_ETNIA`

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_PUEBLO_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_PUEBLO_LISTA SECURE LOOKUP

Replaces the value with one of the 16 values in `pa-pueblos.txt`, chosen by a keyed hash.

Used by: `PA_PUEBLO`

CONFIGURATION

lookupFile	pa-pueblos.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_QUINTIL SECURE LOOKUP

Replaces the value with one of the 5 values in `pa-quintil.txt`, chosen by a keyed hash.

Used by: `PA_NIVEL_SOCIOECONOMICO`

CONFIGURATION

lookupFile	pa-quintil.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_REDACTAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L1_CREDENCIAL`, `PA_L3_BIOMETRICO`, `PA_L3_INFORMACION_GENETICA`

PATTERNS, TRIED IN ORDER

(.+)	each character → X
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_REFERENCIA_CREDITO SECURE LOOKUP

Replaces the value with one of the 10 values in `pa-referencia-credito.txt`, chosen by a keyed hash.

Used by: `PA_APC`

CONFIGURATION

lookupFile	pa-referencia-credito.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_RELIGION REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_RELIGION_LISTA .

Used by: PA_L3_RELIGION

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_RELIGION_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_RELIGION_LISTA SECURE LOOKUP

Replaces the value with one of the 14 values in pa-religiones.txt , chosen by a keyed hash.

Used by: PA_RELIGION

CONFIGURATION

lookupFile	pa-religiones.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_RUC REGEX DECOMPOSE

Tries 4 patterns in order and gives each part of the value its own action; when none matches: PA_CM_ALFANUM .

Used by: PA_L1_RUC

PATTERNS, TRIED IN ORDER

((?:([1-9] 1[0-3])(?:AV PI)?-\d{1,4}-\d{1,6} (?:E N PE)-\d{1,4}-\d{1,6}))(?: +DV + +DV - +)\d{2})	PA_RUC_DV
((?:([1-9] 1[0-3])(?:AV PI)?-\d{1,4}-\d{1,6} (?:E N PE)-\d{1,4}-\d{1,6}))	PA_CEDULA
([1-9] 1[0-3])(-NT-)(\d{1,4})(-)(\d{1,6})(.*)	PA_PROVINCIA_CEDULA · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM · PA_CM_ALFANUM
(\d{3,10}-\d{1,4}-\d{1,7}.*)	keep
otherwise	PA_CM_ALFANUM

CONFIGURATION

trimInput	yes
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requireMask	no
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PA_RUC_DV STRING ALGORITHM CHAIN

Applies PA_RUC_DV_1 → PA_RUC_DV_2 → PA_RUC_DV_3 → PA_RUC_DV_4 → PA_RUC_DV_5 , in this order.

Used by: PA_RUC

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_1
algorithmReferences[2]	PA_RUC_DV_2
algorithmReferences[3]	PA_RUC_DV_3
algorithmReferences[4]	PA_RUC_DV_4
algorithmReferences[5]	PA_RUC_DV_5

PA_RUC_DV_1 STRING ALGORITHM CHAIN

Applies PA_RUC_DV_PREPARAR → PA_RUC_DV_IDA_01 → PA_RUC_DV_IDA_02 → PA_RUC_DV_IDA_03 → PA_RUC_DV_IDA_04 → PA_RUC_DV_IDA_05 → PA_RUC_DV_IDA_06 → PA_RUC_DV_IDA_07 , in this order.

Used by: PA_RUC_DV

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_PREPARAR
algorithmReferences[2]	PA_RUC_DV_IDA_01
algorithmReferences[3]	PA_RUC_DV_IDA_02
algorithmReferences[4]	PA_RUC_DV_IDA_03
algorithmReferences[5]	PA_RUC_DV_IDA_04
algorithmReferences[6]	PA_RUC_DV_IDA_05
algorithmReferences[7]	PA_RUC_DV_IDA_06
algorithmReferences[8]	PA_RUC_DV_IDA_07

PA_RUC_DV_2 STRING ALGORITHM CHAIN

Applies PA_RUC_DV_IDA_08 → PA_RUC_DV_IDA_09 → PA_RUC_DV_IDA_10 → PA_RUC_DV_IDA_11 → PA_RUC_DV_IDA_12 → PA_RUC_DV_IDA_13 → PA_RUC_DV_IDA_14 → PA_RUC_DV_IDA_15 , in this order.

Used by: PA_RUC_DV

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_IDA_08
algorithmReferences[2]	PA_RUC_DV_IDA_09
algorithmReferences[3]	PA_RUC_DV_IDA_10
algorithmReferences[4]	PA_RUC_DV_IDA_11
algorithmReferences[5]	PA_RUC_DV_IDA_12
algorithmReferences[6]	PA_RUC_DV_IDA_13
algorithmReferences[7]	PA_RUC_DV_IDA_14
algorithmReferences[8]	PA_RUC_DV_IDA_15

PA_RUC_DV_3 STRING ALGORITHM CHAIN

Applies PA_RUC_DV_CALCULAR → PA_RUC_DV_VUELTA_01 → PA_RUC_DV_VUELTA_02 → PA_RUC_DV_VUELTA_03 → PA_RUC_DV_VUELTA_04 → PA_RUC_DV_VUELTA_05 → PA_RUC_DV_VUELTA_06 → PA_RUC_DV_VUELTA_07 , in this order.

Used by: PA_RUC_DV

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_CALCULAR
algorithmReferences[2]	PA_RUC_DV_VUELTA_01
algorithmReferences[3]	PA_RUC_DV_VUELTA_02
algorithmReferences[4]	PA_RUC_DV_VUELTA_03
algorithmReferences[5]	PA_RUC_DV_VUELTA_04
algorithmReferences[6]	PA_RUC_DV_VUELTA_05
algorithmReferences[7]	PA_RUC_DV_VUELTA_06
algorithmReferences[8]	PA_RUC_DV_VUELTA_07

PA_RUC_DV_4 STRING ALGORITHM CHAIN

Applies PA_RUC_DV_VUELTA_08 → PA_RUC_DV_VUELTA_09 → PA_RUC_DV_VUELTA_10 → PA_RUC_DV_VUELTA_11 → PA_RUC_DV_VUELTA_12 → PA_RUC_DV_VUELTA_13 → PA_RUC_DV_VUELTA_14 → PA_RUC_DV_VUELTA_15 , in this order.

Used by: PA_RUC_DV

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_VUELTA_08
algorithmReferences[2]	PA_RUC_DV_VUELTA_09
algorithmReferences[3]	PA_RUC_DV_VUELTA_10
algorithmReferences[4]	PA_RUC_DV_VUELTA_11
algorithmReferences[5]	PA_RUC_DV_VUELTA_12
algorithmReferences[6]	PA_RUC_DV_VUELTA_13
algorithmReferences[7]	PA_RUC_DV_VUELTA_14
algorithmReferences[8]	PA_RUC_DV_VUELTA_15

PA_RUC_DV_5 STRING ALGORITHM CHAIN

Applies PA_RUC_DV_VUELTA_16 → PA_RUC_DV_RESTAURAR , in this order.

Used by: PA_RUC_DV

CONFIGURATION

algorithmReferences[1]	PA_RUC_DV_VUELTA_16
algorithmReferences[2]	PA_RUC_DV_RESTAURAR

PA_RUC_DV_CALCULAR REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

(.{3})(.*)	PA_RUC_DV_RESULTADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_01 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)({4})({1})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_02 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)({4})({2})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_03 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)({4})({3})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_04 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)({4})({4})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_05 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)({4})({5})	keep · PA_RUC_DV_PAS0 · keep
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otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no

PA_RUC_DV_IDA_06 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)(.{4})(.{6})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_07 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_1

PATTERNS, TRIED IN ORDER

(.*)(.{4})(.{7})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_08 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)(.{4})(.{8})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_09 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)(.{4})(.{9})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_10 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)({4})({10})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_11 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)({4})({11})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_12 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)({4})({12})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_IDA_13 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_2

PATTERNS, TRIED IN ORDER

(.*)({4})({13})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep

CONFIGURATION

trimInput	yes
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requireMask	no
PA_RUC_DV_IDA_14 REGEX DECOMPOSE	
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
Used by: PA_RUC_DV_2	
PATTERNS, TRIED IN ORDER	
(.*)(.{4})(.{14})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no
PA_RUC_DV_IDA_15 REGEX DECOMPOSE	
Matches one pattern and gives each part of the value its own action; when none matches: keep.	
Used by: PA_RUC_DV_2	
PATTERNS, TRIED IN ORDER	
(.*)(.{4})(.{15})	keep · PA_RUC_DV_PAS0 · keep
otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no
PA_RUC_DV_LETRAS DATA CLEANSING	
Replaces each value found in pa-ruc-dv-letras.txt (5 pairs) by its counterpart; values not in the file stay as they are.	
Used by: PA_RUC_DV_PREPARAR	
CONFIGURATION	
lookupFile	pa-ruc-dv-letras.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes
PA_RUC_DV_LETRAS_DE_VUELTA DATA CLEANSING	
Replaces each value found in pa-ruc-dv-letras-de-vuelta.txt (5 pairs) by its counterpart; values not in the file stay as they are.	
Used by: PA_RUC_DV_RESTAURAR	
CONFIGURATION	
lookupFile	pa-ruc-dv-letras-de-vuelta.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_RUC_DV_PASO DATA CLEANSING

Replaces each value found in `pa-ruc-dv-paso.txt` (32,317 pairs) by its counterpart; values not in the file stay as they are.

Used by: `PA_RUC_DV_IDA_01`, `PA_RUC_DV_IDA_02`, `PA_RUC_DV_IDA_03`, `PA_RUC_DV_IDA_04`, `PA_RUC_DV_IDA_05`, `PA_RUC_DV_IDA_06`, `PA_RUC_DV_IDA_07`, `PA_RUC_DV_IDA_08`, `PA_RUC_DV_IDA_09`, `PA_RUC_DV_IDA_10`, `PA_RUC_DV_IDA_11`, `PA_RUC_DV_IDA_12`, `PA_RUC_DV_IDA_13`, `PA_RUC_DV_IDA_14`, `PA_RUC_DV_IDA_15`

CONFIGURATION

lookupFile	pa-ruc-dv-paso.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_RUC_DV_PREPARAR REGEX DECOMPOSE

Tries 16 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: `PA_RUC_DV_1`

PATTERNS, TRIED IN ORDER

<code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})(+DV +)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove
<code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})(+DV +)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove
<code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})(+DV +)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove
<code>(N PE)(-)(\d{1,4})(-)(\d{1,6})(+DV +)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a!</code> · remove
<code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})(+DV)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove
<code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})(+DV)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove
<code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})(+DV)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove
<code>(N PE)(-)(\d{1,4})(-)(\d{1,6})(+DV)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a#</code> · remove
<code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove
<code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove
<code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})(-)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove
<code>(N PE)(-)(\d{1,4})(-)(\d{1,6})(-)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a\$</code> · remove
<code>([1-9] 1[0-3])(-)(\d{1,4})(-)(\d{1,6})(+)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove
<code>([1-9] 1[0-3])(AV PI)(-)(\d{1,4})(-)(\d{1,6})(+)(\d{2})</code>	<code>PA_PROVINCIA_CEDULA</code> · <code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove
<code>(E)(-)([1-9] 1[0-3])(-)(\d{1,6})(+)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_PROVINCIA_CEDULA</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove
<code>(N PE)(-)(\d{1,4})(-)(\d{1,6})(+)(\d{2})</code>	<code>PA_RUC_DV_LETRAS</code> · keep · <code>PA_CM_ALFANUM</code> · replace with <code>_</code> · <code>PA_CM_ALFANUM</code> · replace with <code>00a%</code> · remove
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_RESTAURAR REGEX DECOMPOSE

Tries 12 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_5

PATTERNS, TRIED IN ORDER

(\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)	keep · keep · keep · replace with - · keep · replace with DV · keep
(\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)	keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep
([enq])(-)(\d{1,4})(_)(\d{1,6})(!)(\d\d)	PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep
(\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)	keep · keep · keep · replace with - · keep · replace with DV · keep
(\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)	keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep
([enq])(-)(\d{1,4})(_)(\d{1,6})(#)(\d\d)	PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with DV · keep
(\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d)	keep · keep · keep · replace with - · keep · replace with - · keep
(\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d)	keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with - · keep
([enq])(-)(\d{1,4})(_)(\d{1,6})(\$)(\d\d)	PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with - · keep
(\d{1,2})(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)	keep · keep · keep · replace with - · keep · replace with · keep
(\d{1,2})([vi])(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)	keep · PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with · keep
([enq])(-)(\d{1,4})(_)(\d{1,6})(%)(\d\d)	PA_RUC_DV_LETRAS_DE_VUELTA · keep · keep · replace with - · keep · replace with · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_RESULTADO DATA CLEANSING

Replaces each value found in pa-ruc-dv-resultado.txt (1,815 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA_RUC_DV_CALCULAR

CONFIGURATION

lookupFile	pa-ruc-dv-resultado.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_RUC_DV TRASLADO DATA CLEANSING

Replaces each value found in pa-ruc-dv-traslado.txt (2,100 pairs) by its counterpart; values not in the file stay as they are.

Used by: PA_RUC_DV_VUELTA_01 , PA_RUC_DV_VUELTA_02 , PA_RUC_DV_VUELTA_03 , PA_RUC_DV_VUELTA_04 , PA_RUC_DV_VUELTA_05 , PA_RUC_DV_VUELTA_06 , PA_RUC_DV_VUELTA_07 , PA_RUC_DV_VUELTA_08 , PA_RUC_DV_VUELTA_09 , PA_RUC_DV_VUELTA_10 ,

PA_RUC_DV_VUELTA_11 , PA_RUC_DV_VUELTA_12 , PA_RUC_DV_VUELTA_13 , PA_RUC_DV_VUELTA_14 , PA_RUC_DV_VUELTA_15 , PA_RUC_DV_VUELTA_16

CONFIGURATION

lookupFile	pa-ruc-dv-traslado.txt
delimiter	,
caseSensitive	yes
trimWhitespace	yes

PA_RUC_DV_VUELTA_01 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

(.{0})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_02 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

(.{1})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_03 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

(.{2})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_04 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

(.{3})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
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otherwise	keep
CONFIGURATION	
trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_05 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

<code>(.{4})(.{3})(.*)</code>	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_06 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

<code>(.{5})(.{3})(.*)</code>	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_07 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_3

PATTERNS, TRIED IN ORDER

<code>(.{6})(.{3})(.*)</code>	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_08 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

<code>(.{7})(.{3})(.*)</code>	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_09 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{8})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_10 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{9})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_11 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{10})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_12 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{11})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
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requireMask	no
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PA_RUC_DV_VUELTA_13 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{12})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_14 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{13})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_15 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_4

PATTERNS, TRIED IN ORDER

(.{14})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_RUC_DV_VUELTA_16 REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: PA_RUC_DV_5

PATTERNS, TRIED IN ORDER

(.{15})(.{3})(.*)	keep · PA_RUC_DV TRASLADO · keep
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_SEGURO_SOCIAL REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches:

PA_SEGURO_SOCIAL_LISTA .

Used by: PA_FIN_SEGURO_SOCIAL

PATTERNS, TRIED IN ORDER

(?i)(s[ii] no true false [sny01])	PA_BANDERA
(\d{1,2})	PA_CM_ALFANUM
otherwise	PA_SEGURO_SOCIAL_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_SEGURO_SOCIAL_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in pa-seguro-social.txt , chosen by a keyed hash.

Used by: PA_SEGURO_SOCIAL

CONFIGURATION

lookupFile	pa-seguro-social.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SEXO REGEX DECOMPOSE

Tries 5 patterns in order and gives each part of the value its own action; when none matches: keep.

Used by: PA_L2_SEX0

PATTERNS, TRIED IN ORDER

(?i)(femenino masculino)	PA_SEX0_PALABRA
(?i)(mujer hombre)	PA_SEX0_MUJER_HOMBRE
(?i)(female male)	PA_SEX0_INGLES
(?i)([fm])	PA_SEX0_F_M
(?i)([hm])	PA_SEX0_H_M
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_SEX0_F_M SECURE LOOKUP

Replaces the value with one of the 2 values in pa-sexo-f-m.txt , chosen by a keyed hash.

Used by: PA_SEX0

CONFIGURATION

lookupFile	pa-sexo-f-m.txt
hashMethod	SHA256

maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SEXO_H_M SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-h-m.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

CONFIGURATION

lookupFile	pa-sexo-h-m.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SEXO_INGLES SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-ingles.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

CONFIGURATION

lookupFile	pa-sexo-ingles.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SEXO_MUJER_HOMBRE SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-mujer-hombre.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

CONFIGURATION

lookupFile	pa-sexo-mujer-hombre.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SEXO_PALABRA SECURE LOOKUP

Replaces the value with one of the 2 values in `pa-sexo-palabra.txt` , chosen by a keyed hash.

Used by: `PA_SEX0`

CONFIGURATION

lookupFile	pa-sexo-palabra.txt
hashMethod	SHA256

maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_SUPRIMIR SECURE LOOKUP

Replaces the value with one of the 1 values in `pa-sin-informacion.txt` , chosen by a keyed hash.

Used by: `PA_CATEGORIA_SUPRIMIDA`

CONFIGURATION

lookupFile	pa-sin-informacion.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_LOOKUP_FILE
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

PA_TARJETA REGEX DECOMPOSE

Matches one pattern and gives each part of the value its own action; when none matches: keep.

Used by: `PA_L1_TARJETA`

PATTERNS, TRIED IN ORDER

<code>(\d[\d\s\~]{11,22}\d)</code>	<code>PA_TARJETA_LUHN</code>
otherwise	keep

CONFIGURATION

trimInput	yes
requireMask	no

PA_TARJETA_LUHN PAYMENT CARD

Replaces the card number with another that passes the Luhn check.

Used by: `PA_TARJETA`

CONFIGURATION

minMaskedPositions	6
preserve	0

PA_TELEFONO REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: `PA_CM_ALFANUM` .

Used by: `PA_L1_TELEFONO`

PATTERNS, TRIED IN ORDER

<code>(\+?(?07\)?[\s\~]?)?(6)(\d{3})([\s\~]?) (\d{4})</code>	<code>keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM</code>
<code>(\+?(?07\)?[\s\~]?)?([2-9])(\d{2}) ([\s\~]?) (\d{4})</code>	<code>keep · keep · PA_CM_ALFANUM · keep · PA_CM_ALFANUM</code>
otherwise	<code>PA_CM_ALFANUM</code>

CONFIGURATION

trimInput	yes
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requireMask	no
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PA_TEXTO_LIBRE FREE TEXT REDACTION

Replaces the text matching 7 patterns with [DATO] , and the 1,094 terms of pa-texto-nombres.txt with [NOMBRE] .

Used by: PA_L1_TEXTO_LIBRE , PA_L3_SALUD_TEXTO_CLINICO

PATTERNS, TRIED IN ORDER

(?<![\w\~])(([1-9] 1[0-3])(AV PI)? E N PE)-\d{1,4}-\d{1,6}(?![\w\~])
[\w.+~]+@[~\w\~]+(\. [\w\~]+)+
(?<!\d)(\+?507)?6\d{3}-?\d{4}(?! \d)
(?<![\d\~]) [2-9]\d{2}-\d{4}(?! [\d\~])
(?<!\d)\d{13,19}(?! \d)
(?<![A-Za-z]) [A-Z]{2}\d{4}(?! [A-Za-z\d])
(?<!\d)(\d{1,3}\.){3}\d{1,3}(?! \d)

CONFIGURATION

regExRedactValue	[DATO]
lookupFile	pa-texto-nombres.txt
lookupFileRedactValue	[NOMBRE]
isDenyList	yes

PA_VIVIENDA REGEX DECOMPOSE

Tries 2 patterns in order and gives each part of the value its own action; when none matches: PA_VIVIENDA_LISTA .

Used by: PA_FIN_VIVIENDA

PATTERNS, TRIED IN ORDER

(?i)(s[ií] no true false [sny01])	PA_BANDERA
(\d{1,6})	PA_CM_ALFANUM
otherwise	PA_VIVIENDA_LISTA

CONFIGURATION

trimInput	yes
requireMask	no

PA_VIVIENDA_LISTA SECURE LOOKUP

Replaces the value with one of the 6 values in pa-vivienda.txt , chosen by a keyed hash.

Used by: PA_VIVIENDA

CONFIGURATION

lookupFile	pa-vivienda.txt
hashMethod	SHA256
maskedValueCase	PRESERVE_INPUT
inputCaseSensitive	no
trimWhitespaceFromInput	yes
trimWhitespaceInLookupFile	yes

APPENDIX B

Files

The lookup and detection files the set reads. On a Masking Engine each one is uploaded with the object that reads it — the lookup file of a Secure Lookup or Data Cleansing algorithm, the value list of a List classifier — and referenced by that object's configuration.

FILE	LINES	USED BY	FIRST LINES
pa- antecedentes.txt	3	PA_ANTECEDENTES_LIST A	Sin antecedentes penales · Con antecedentes penales · Sin información
pa-apellidos.txt	200	PA_APELLIDO	Rodríguez · González · Sánchez · Martínez
pa-bandera-0- 1.txt	2	PA_BANDERA_0_1	0 · 1
pa-bandera-s- n.txt	2	PA_BANDERA_S_N	S · N
pa-bandera-si- no.txt	2	PA_BANDERA_SI_NO	Sí · No
pa-bandera-true- false.txt	2	PA_BANDERA_TRUE_FALS E	true · false
pa-bandera-y- n.txt	2	PA_BANDERA_Y_N	Y · N
pa-barriadas.txt	75	PA_BARRIADA	Villa Lucre · Condado del Rey · Brisas del Golf · Altos de Panamá
pa-cie10- decimal.txt	33	PA_CIE10_DECIMAL	E11.9 · E78.5 · J45.9 · J06.9
pa-cie10.txt	37	PA_CIE10	I10 · E11 · E78 · J45
pa-codigo- corregimiento- generalizado.txt	1,331	PA_CODIGO_CORREGIMIEN TO	010101,010206 · 10101,10206 · 010102,010206 · 10102,10206
pa-codigo- distrito- generalizado.txt	148	PA_CODIGO_DISTRITO	0101,0104 · 101,104 · 0102,0102 · 102,102
pa- corregimiento- generalizado.txt	1,760	PA_CORREGIMIENTO	Bocas del Toro El Empalme · BOCAS DEL TORO EL EMPALME · Bastimentos El Empalme · BASTIMENTOS EL EMPALME
pa-decil- alto.txt	5	PA_DECIL_ALTO	6 · 7 · 8 · 9
pa-decil.txt	10	PA_DECIL	1 · 2 · 3 · 4
pa-detectar- apellidos.txt	258	PA_L1_APELLIDO – List PA_L1_NOMBRE_COMPLETO – List	Rodríguez · Rodriguez · González · Gonzalez
pa-detectar- bandera.txt	8	PA_L3_ETNIA – List	s · n · sí · si
pa-detectar- codigo-lugar.txt	1,405	PA_L2_CODIGO_LUGAR – List	010101 · 10101 · 010102 · 10102

FILE	LINES	USED BY	FIRST LINES
pa-detectar-corregimientos.txt	814	PA_L2_CORREGIMIENTO – List	Bocas del Toro · Bastimentos · Punta Laurel · Tierra Oscura
pa-detectar-discapacidad.txt	26	PA_L3_SALUD_DISCAPACIDAD – List	Física · Fisica · Visual · Auditiva
pa-detectar-distritos.txt	111	PA_L2_DISTRITO – List	Bocas del Toro · Changuinola · Chiriquí Grande · Chiriqui Grande
pa-detectar-escolaridad.txt	28	PA_L2_ESCOLARIDAD – List	Ningún grado · Ningun grado · Primaria incompleta · Primaria completa
pa-detectar-estado-civil.txt	18	PA_L2_ESTADO_CIVIL – List	soltero · soltera · casado · casada
pa-detectar-grupo-sanguineo.txt	26	PA_L3_SALUD_GRUPO_SANGUINEO – List	A+ · A- · B+ · B-
pa-detectar-identidad-genero.txt	17	PA_L3_IDENTIDAD_GENERO – List	Mujer · Hombre · Mujer trans · Hombre trans
pa-detectar-lenguas.txt	32	PA_L3 LENGUA_INDIGENA – List	Ngäbere · Ngabere · Buglere · Guna
pa-detectar-medicamentos.txt	58	PA_L3_SALUD_MEDICAMENTO – List	Acetaminofén · Acetaminofen · Ibuprofeno · Naproxeno
pa-detectar-nacionalidades.txt	63	PA_L2_NACIONALIDAD – List	Panameña · Panamena · Colombiana · Venezolana
pa-detectar-nombres.txt	291	PA_L1_NOMBRE – List PA_L1_NOMBRE_COMPLETO – List	María · Maria · José · Jose
pa-detectar-opiniones-politicas.txt	14	PA_L3_OPINION_POLITICA – List	Izquierda · Centro izquierda · Centro · Centro derecha
pa-detectar-orientacion-sexual.txt	17	PA_L3_ORIENTACION_SEXUAL – List	Heterosexual · Homosexual · Lesbiana · Gay
pa-detectar-partidos.txt	35	PA_L3_AFILIACION POLITICA – List	Partido Revolucionario Democrático · Partido Revolucionario Democratico · Realizando Metas · Cambio Democrático
pa-detectar-pobreza.txt	10	PA_FIN_NIVEL_SOCIOECONOMICO – List	pobreza extrema · pobreza general · pobre · no pobre
pa-detectar-programas.txt	23	PA_FIN_PROGRAMA_SOCIAL – List	120 a los 65 · Red de Oportunidades · Ángel Guardián · Angel Guardian
pa-detectar-pueblos.txt	52	PA_L3_ETNIA – List	Guna · Ngäbe · Ngabe · Buglé
pa-detectar-religiones.txt	61	PA_L3_RELIGION – List	Católica · Catolica · Evangélica · Evangelica
pa-detectar-seguro-social.txt	20	PA_FIN_SEGURO_SOCIAL – List	Asegurado directo · Beneficiario · Jubilado o pensionado por vejez · Pensionado por enfermedad o accidente

FILE	LINES	USED BY	FIRST LINES
pa-detectar-sexo.txt	11	PA_L2_SEX0 – List	h · m · f · hombre
pa-detectar-sindicatos.txt	15	PA_L3_AFILIACION_SINDICAL – List	SUNTRACS · CONATO · CONUSI · CONATLAB
pa-detectar-vivienda.txt	17	PA_FIN_VIVIENDA – List	Propia · Propia hipotecada · Alquilada · Cedida o prestada
pa-diagnosticos.txt	30	PA_DIAGNOSTICO_TEXTO	Hipertensión arterial · Diabetes mellitus tipo 2 · Asma bronquial · Lumbalgia
pa-direcciones.txt	3,000	PA_DIRECCION	Barriada Villa del Carmen, Calle 5 D, Casa 389 · Don Bosco, Calle 14 B, Casa 147 · Calle 63, Edificio Torre del Mar, Piso 10, Apto. 16E · Villa Grecia, Casa No. 46
pa-discapacidad.txt	8	PA_DISCAPACIDAD_LISTA	Física · Visual · Auditiva · Intelectual
pa-distrito-generalizado.txt	234	PA_DISTRITO	Bocas del Toro Almirante · BOCAS DEL TORO ALMIRANTE · Changuinola Changuinola · CHANGUINOLA CHANGUINOLA
pa-empleadores.txt	25	PA_EMPLEADOR	Comercializadora del Istmo, S.A. · Servicios Integrales del Pacífico, S.A. · Constructora Cerro Ancón, S.A. · Transportes Canal Seguro, S.A.
pa-escolaridad.txt	13	PA_ESCOLARIDAD	Ningún grado · Primaria incompleta · Primaria completa · Premedia incompleta
pa-escuelas.txt	15	PA_ESCUELA	Escuela República de Colombia · Escuela Juan Demóstenes Arosemena · Centro Básico General Las Mañanitas · Centro de Educación Básica General Guararé
pa-estado-civil.txt	7	PA_ESTADO_CIVIL	Unido · Casado · Soltero · Separado de matrimonio
pa-grupo-sanguineo.txt	8	PA_GRUPO_SANGUINEO	A+ · A- · B+ · B-
pa-identidad-genero.txt	8	PA_IDENTIDAD_GENERO_LISTA	Mujer · Hombre · Mujer trans · Hombre trans
pa-lenguas.txt	9	PA LENGUA_INDIGENA_LISTA	Ngäbere · Buglere · Guna · Emberá
pa-lugares-poblados.txt	52	PA_LUGAR_POBLADO	Penonomé · Cativá · Sabanitas · San Juan
pa-medicamentos.txt	26	PA_MEDICAMENTO_LISTA	Acetaminofén · Ibuprofeno · Naproxeno · Diclofenaco
pa-nacionalidades.txt	20	PA_NACIONALIDAD	Panameña · Colombiana · Venezolana · Nicaragüense
pa-nombres.txt	253	PA_NOMBRE_PILA	María · José · Luis · Carlos
pa-ocupaciones.txt	40	PA_OCUPACION	Oficinista · Comerciante · Vendedor · Obrero de la construcción
pa-opiniones-politicas.txt	7	PA_OPINION_POLITICA_LISTA	Izquierda · Centro izquierda · Centro · Centro derecha
pa-orientacion-sexual.txt	9	PA_ORIENTACION_SEXUAL_LISTA	Heterosexual · Homosexual · Lesbiana · Gay
pa-partidos.txt	11	PA_PARTIDO_POLITICO_LISTA	Partido Revolucionario Democrático · Realizando Metas · Cambio Democrático · Partido Panameñista

FILE	LINES	USED BY	FIRST LINES
pa-pobreza.txt	4	PA_POBREZA	Pobreza extrema · Pobreza general · No pobre · Vulnerable
pa-programas-sociales.txt	9	PA_PROGRAMA_SOCIAL_LISTA	120 a los 65 · Red de Oportunidades · Ángel Guardián · Bono Alimenticio Nutricional
pa-pueblos.txt	16	PA_PUEBLO_LISTA	Guna · Ngäbe · Buglé · Naso Tjërdi
pa-quintil.txt	5	PA_QUINTIL	1 · 2 · 3 · 4
pa-referencia-credito.txt	10	PA_REFERENCIA_CREDITO	Al día · Morosidad de 30 días · Morosidad de 60 días · Morosidad de 90 días
pa-religiones.txt	14	PA_RELIGION_LISTA	Católica · Evangélica · Adventista del Séptimo Día · Testigo de Jehová
pa-ruc-dv-letras-de-vuelta.txt	5	PA_RUC_DV_LETRAS_DE_VUELTA	v,AV · i,PI · e,E · n,N
pa-ruc-dv-letras.txt	5	PA_RUC_DV_LETRAS	AV,v · PI,i · E,e · N,n
pa-ruc-dv-paso.txt	32,317	PA_RUC_DV_PASO	000a,00b0 · 100a,21b1 · 200a,42b2 · 300a,63b3
pa-ruc-dv-resultado.txt	1,815	PA_RUC_DV_RESULTADO	A8J,81 · 40J,33 · 41J,32 · 4Au,99
pa-ruc-dv-traslado.txt	2,100	PA_RUC_DV TRASLADO	000,000 · 001,100 · 002,200 · 003,300
pa-seguro-social.txt	6	PA_SEGURO_SOCIAL_LISTA	Asegurado directo · Beneficiario · Jubilado o pensionado por vejez · Pensionado por enfermedad o accidente
pa-sexo-f-m.txt	2	PA_SEX0_F_M	F · M
pa-sexo-h-m.txt	2	PA_SEX0_H_M	H · M
pa-sexo-ingles.txt	2	PA_SEX0_INGLES	Female · Male
pa-sexo-mujer-hombre.txt	2	PA_SEX0_MUJER_HOMBRE	Mujer · Hombre
pa-sexo-palabra.txt	2	PA_SEX0_PALABRA	Femenino · Masculino
pa-sin-informacion.txt	1	PA_SUPRIMIR	Sin información
pa-texto-nombres.txt	1,094	PA_TEXTO_LIBRE	María · MARÍA · Maria · MARIA
pa-vivienda.txt	6	PA_VIVIENDA_LISTA	Propia · Propia hipotecada · Alquilada · Cedida o prestada

Exclusion lists

Words that keep a column-name rule from matching when they follow it — so that `NOMBRE_PRODUCTO` is not taken for a person's name.

E1 producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto cliente cte cli paciente pac trabajador trab empleado colaborador funcionario servidor_? publico alumno estudiante afiliado asegurado cotizante jubilado pensionado beneficiario benef titular madre padre acudiente tutor conyuge pareja medico doctor profesional responsable contacto emergencia fiador codeudor deudor remitente destinatario receptor firmante testigo representante apoderado propietario arrendatario inquilino contratante victima denunciante imputado socio donante heredero persona pers conductor chofer vendedor ejecutivo agente solicitante ciudadano elector votante contribuyente jefe_?(de_?)?hogar hijo hija familiar dependiente completos? y_?apellidos? apellidos?

E2 producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto cliente cte cli paciente pac trabajador trab empleado colaborador funcionario servidor_? publico alumno estudiante afiliado asegurado cotizante jubilado pensionado beneficiario benef titular madre padre acudiente tutor conyuge pareja medico doctor profesional responsable contacto emergencia fiador codeudor deudor remitente destinatario receptor firmante testigo representante apoderado propietario arrendatario inquilino contratante victima denunciante imputado socio donante heredero persona pers conductor chofer vendedor ejecutivo agente solicitante ciudadano elector votante contribuyente jefe_?(de_?)?hogar hijo hija familiar dependiente completo compl full y_? apellidos? apellidos?

-
- E3** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto completo
-
- E4** producto prod articulo art item empresa emp compania razon comercial fantasia archivo arch file calle via avenida barriada urbanizacion corregimiento correg distrito dist provincia prov comarca lugar_?poblado region ciudad pais banco sucursal plaza plan campana proyecto servicio equipo tabla columna campo usuario user host servidor dominio marca modelo categoria tipo colegio escuela centro_?educativo establecimiento institucion curso carrera materia documento doc centro unidad policlinica clinica hospital area depto departamento puesto sistema app aplicacion grupo lista reporte informe proceso evento tarea perfil rol clase objeto cuenta medicamento farmaco diagnostico estudio programa beneficio partido sindicato religion etnia pueblo lengua lugar local tienda almacen bodega proveedor convenio contrato poliza seguro aseguradora org organizacion dependencia entidad parametro variable metodo funcion indice job script cola recurso red dispositivo plantilla template imagen foto pagina sitio modulo menu opcion accion permiso concepto impuesto moneda emisor comprobante factura pago forma zona ruta linea giro actividad rama sector finca edificio ph proyecto
-
- E5** ip mac web url correo mail electronica email
-
- E6** producto prod articulo sucursal interna tipo registro primaria foranea social banco pais moneda catastral ubicacion zona area postal
-
- E7** fijo variable monto adicional servicio tarifa valor cuenta extra automatico recurrente
-
- E8** medicos patologicos personales_?patologicos heredofamiliares gineco obstetricos familiares clinicos academicos laborales crediticios comerciales generales
-

APPENDIX D

Profile set and classifiers

The profile set `PA - Ley 81 de 2019 - v2` runs these 135 classifiers, with an assignment threshold of 60%.

Identificadores directos, cuasi-identificadores, datos sensibles (art. 4, 11), datos económicos y de crédito, y antecedentes penales, conforme a la Ley 81 de 2019 y el Decreto Ejecutivo 285 de 2021.

CLASSIFIER	KIND	DOMAIN
PA_L1_CEDULA - Path	COLUMN NAME	PA_L1_CEDULA
PA_L1_CEDULA - Type	DATA TYPE	PA_L1_CEDULA
PA_L1_CEDULA - Regex	VALUE FORMAT	PA_L1_CEDULA
PA_L1_RUC - Path	COLUMN NAME	PA_L1_RUC
PA_L1_RUC - Type	DATA TYPE	PA_L1_RUC
PA_L1_RUC - Regex	VALUE FORMAT	PA_L1_RUC
PA_L1_RUC_DV - Path	COLUMN NAME	PA_L1_RUC_DV
PA_L1_RUC_DV - Type	DATA TYPE	PA_L1_RUC_DV
PA_L1_PASAPORTE - Path	COLUMN NAME	PA_L1_PASAPORTE
PA_L1_PASAPORTE - Regex	VALUE FORMAT	PA_L1_PASAPORTE
PA_L1_DOCUMENTO_MIGRATORIO - Path	COLUMN NAME	PA_L1_DOCUMENTO_MIGRATORIO
PA_L1_SEGURO_SOCIAL - Path	COLUMN NAME	PA_L1_SEGURO_SOCIAL
PA_L1_IDONEIDAD - Path	COLUMN NAME	PA_L1_IDONEIDAD
PA_L1_NOMBRE - Path	COLUMN NAME	PA_L1_NOMBRE
PA_L1_NOMBRE - List	KNOWN VALUES	PA_L1_NOMBRE
PA_L1_APELLIDO - Path	COLUMN NAME	PA_L1_APELLIDO
PA_L1_APELLIDO - List	KNOWN VALUES	PA_L1_APELLIDO
PA_L1_NOMBRE_COMPLETO - Path	COLUMN NAME	PA_L1_NOMBRE_COMPLETO
PA_L1_NOMBRE_COMPLETO - List	KNOWN VALUES	PA_L1_NOMBRE_COMPLETO
PA_L1_EMAIL - Path	COLUMN NAME	PA_L1_EMAIL
PA_L1_EMAIL - Type	DATA TYPE	PA_L1_EMAIL
PA_L1_EMAIL - Regex	VALUE FORMAT	PA_L1_EMAIL
PA_L1_TELEFONO - Path	COLUMN NAME	PA_L1_TELEFONO
PA_L1_TELEFONO - Regex	VALUE FORMAT	PA_L1_TELEFONO
PA_L1_DIRECCION - Path	COLUMN NAME	PA_L1_DIRECCION
PA_L1_DIRECCION - Type	DATA TYPE	PA_L1_DIRECCION
PA_L1_DIRECCION - Regex	VALUE FORMAT	PA_L1_DIRECCION
PA_L1_DIRECCION_COMPLEMENTO - Path	COLUMN NAME	PA_L1_DIRECCION_COMPLEMENTO
PA_L1_CUENTA_BANCARIA - Path	COLUMN NAME	PA_L1_CUENTA_BANCARIA
PA_L1_TARJETA - Path	COLUMN NAME	PA_L1_TARJETA
PA_L1_TARJETA - Regex	VALUE FORMAT	PA_L1_TARJETA

CLASSIFIER	KIND	DOMAIN
PA_L1_IP – Path	COLUMN NAME	PA_L1_IP
PA_L1_IP – Regex	VALUE FORMAT	PA_L1_IP
PA_L1_DISPPOSITIVO – Path	COLUMN NAME	PA_L1_DISPPOSITIVO
PA_L1_DISPPOSITIVO – Regex	VALUE FORMAT	PA_L1_DISPPOSITIVO
PA_L1_PLACA – Path	COLUMN NAME	PA_L1_PLACA
PA_L1_PLACA – Regex	VALUE FORMAT	PA_L1_PLACA
PA_L1_VIN – Path	COLUMN NAME	PA_L1_VIN
PA_L1_VIN – Regex	VALUE FORMAT	PA_L1_VIN
PA_L1_FINCA – Path	COLUMN NAME	PA_L1_FINCA
PA_L1_NUMERO_CONTRATO – Path	COLUMN NAME	PA_L1_NUMERO_CONTRATO
PA_L1_USUARIO_RRSS – Path	COLUMN NAME	PA_L1_USUARIO_RRSS
PA_L1_CREDENCIAL – Path	COLUMN NAME	PA_L1_CREDENCIAL
PA_L1_GEOLOCALIZACION – Path	COLUMN NAME	PA_L1_GEOLOCALIZACION
PA_L1_GEOLOCALIZACION – Regex	VALUE FORMAT	PA_L1_GEOLOCALIZACION
PA_L1_TEXTO_LIBRE – Path	COLUMN NAME	PA_L1_TEXTO_LIBRE
PA_L1_TEXTO_LIBRE – Type	DATA TYPE	PA_L1_TEXTO_LIBRE
PA_L1_TEXTO_LIBRE – Regex	VALUE FORMAT	PA_L1_TEXTO_LIBRE
PA_L2_FECHA_NACIMIENTO – Path	COLUMN NAME	PA_L2_FECHA_NACIMIENTO
PA_L2_FECHA_NACIMIENTO – Type	DATA TYPE	PA_L2_FECHA_NACIMIENTO
PA_L2_ANIO_NACIMIENTO – Path	COLUMN NAME	PA_L2_ANIO_NACIMIENTO
PA_L2_ANIO_NACIMIENTO – Type	DATA TYPE	PA_L2_ANIO_NACIMIENTO
PA_L2_EDAD – Path	COLUMN NAME	PA_L2_EDAD
PA_L2_EDAD – Type	DATA TYPE	PA_L2_EDAD
PA_L2_FECHA_EVENTO – Path	COLUMN NAME	PA_L2_FECHA_EVENTO
PA_L2_FECHA_EVENTO – Type	DATA TYPE	PA_L2_FECHA_EVENTO
PA_L2_SEX0 – Path	COLUMN NAME	PA_L2_SEX0
PA_L2_SEX0 – List	KNOWN VALUES	PA_L2_SEX0
PA_L2_DISTRITO – Path	COLUMN NAME	PA_L2_DISTRITO
PA_L2_DISTRITO – List	KNOWN VALUES	PA_L2_DISTRITO
PA_L2_CORREGIMIENTO – Path	COLUMN NAME	PA_L2_CORREGIMIENTO
PA_L2_CORREGIMIENTO – List	KNOWN VALUES	PA_L2_CORREGIMIENTO
PA_L2_CODIGO_LUGAR – Path	COLUMN NAME	PA_L2_CODIGO_LUGAR
PA_L2_CODIGO_LUGAR – Type	DATA TYPE	PA_L2_CODIGO_LUGAR
PA_L2_CODIGO_LUGAR – List	KNOWN VALUES	PA_L2_CODIGO_LUGAR
PA_L2_LUGAR_POBLADO – Path	COLUMN NAME	PA_L2_LUGAR_POBLADO
PA_L2_BARRIADA – Path	COLUMN NAME	PA_L2_BARRIADA
PA_L2_CODIGO_POSTAL – Path	COLUMN NAME	PA_L2_CODIGO_POSTAL

CLASSIFIER	KIND	DOMAIN
PA_L2_CODIGO_POSTAL - Regex	VALUE FORMAT	PA_L2_CODIGO_POSTAL
PA_L2_NACIONALIDAD - Path	COLUMN NAME	PA_L2_NACIONALIDAD
PA_L2_NACIONALIDAD - List	KNOWN VALUES	PA_L2_NACIONALIDAD
PA_L2_ESTADO_CIVIL - Path	COLUMN NAME	PA_L2_ESTADO_CIVIL
PA_L2_ESTADO_CIVIL - List	KNOWN VALUES	PA_L2_ESTADO_CIVIL
PA_L2_OCUPACION - Path	COLUMN NAME	PA_L2_OCUPACION
PA_L2_EMPLEADOR - Path	COLUMN NAME	PA_L2_EMPLEADOR
PA_L2_ESCOLARIDAD - Path	COLUMN NAME	PA_L2_ESCOLARIDAD
PA_L2_ESCOLARIDAD - List	KNOWN VALUES	PA_L2_ESCOLARIDAD
PA_L2_ESCUELA - Path	COLUMN NAME	PA_L2_ESCUELA
PA_L2_DEPENDIENTES - Path	COLUMN NAME	PA_L2_DEPENDIENTES
PA_L2_DEPENDIENTES - Type	DATA TYPE	PA_L2_DEPENDIENTES
PA_L3_ETNIA - Path	COLUMN NAME	PA_L3_ETNIA
PA_L3_ETNIA - List	KNOWN VALUES	PA_L3_ETNIA
PA_L3 LENGUA_INDIGENA - Path	COLUMN NAME	PA_L3 LENGUA_INDIGENA
PA_L3 LENGUA_INDIGENA - List	KNOWN VALUES	PA_L3 LENGUA_INDIGENA
PA_L3_SALUD_DIAGNOSTICO - Path	COLUMN NAME	PA_L3_SALUD_DIAGNOSTICO
PA_L3_SALUD_DIAGNOSTICO - Regex	VALUE FORMAT	PA_L3_SALUD_DIAGNOSTICO
PA_L3_SALUD_MEDICAMENTO - Path	COLUMN NAME	PA_L3_SALUD_MEDICAMENTO
PA_L3_SALUD_MEDICAMENTO - List	KNOWN VALUES	PA_L3_SALUD_MEDICAMENTO
PA_L3_SALUD_IDENTIFICADOR - Path	COLUMN NAME	PA_L3_SALUD_IDENTIFICADOR
PA_L3_SALUD_TEXTO_CLINICO - Path	COLUMN NAME	PA_L3_SALUD_TEXTO_CLINICO
PA_L3_SALUD_TEXTO_CLINICO - Type	DATA TYPE	PA_L3_SALUD_TEXTO_CLINICO
PA_L3_SALUD_RESULTADO_EXAMEN - Path	COLUMN NAME	PA_L3_SALUD_RESULTADO_EXAMEN
PA_L3_SALUD_GRUPO_SANGUINEO - Path	COLUMN NAME	PA_L3_SALUD_GRUPO_SANGUINEO
PA_L3_SALUD_GRUPO_SANGUINEO - List	KNOWN VALUES	PA_L3_SALUD_GRUPO_SANGUINEO
PA_L3_SALUD_DISCAPACIDAD - Path	COLUMN NAME	PA_L3_SALUD_DISCAPACIDAD
PA_L3_SALUD_DISCAPACIDAD - List	KNOWN VALUES	PA_L3_SALUD_DISCAPACIDAD
PA_L3_SALUD_SEXUAL_REPRODUCTIVA - Path	COLUMN NAME	PA_L3_SALUD_SEXUAL_REPRODUCTIVA
PA_L3_SALUD_MENTAL_ADICCIONES - Path	COLUMN NAME	PA_L3_SALUD_MENTAL_ADICCIONES
PA_L3_BIOMETRICO - Path	COLUMN NAME	PA_L3_BIOMETRICO
PA_L3_BIOMETRICO - Type	DATA TYPE	PA_L3_BIOMETRICO
PA_L3_INFORMACION_GENETICA - Path	COLUMN NAME	PA_L3_INFORMACION_GENETICA
PA_L3_ORIENTACION_SEXUAL - Path	COLUMN NAME	PA_L3_ORIENTACION_SEXUAL
PA_L3_ORIENTACION_SEXUAL - List	KNOWN VALUES	PA_L3_ORIENTACION_SEXUAL
PA_L3_IDENTIDAD_GENERO - Path	COLUMN NAME	PA_L3_IDENTIDAD_GENERO
PA_L3_IDENTIDAD_GENERO - List	KNOWN VALUES	PA_L3_IDENTIDAD_GENERO

CLASSIFIER	KIND	DOMAIN
PA_L3_VIDA_SEXUAL - Path	COLUMN NAME	PA_L3_VIDA_SEXUAL
PA_L3_RELIGION - Path	COLUMN NAME	PA_L3_RELIGION
PA_L3_RELIGION - List	KNOWN VALUES	PA_L3_RELIGION
PA_L3_CREENCIA_FILOSOFICA_MORAL - Path	COLUMN NAME	PA_L3_CREENCIA_FILOSOFICA_MORAL
PA_L3_OPINION_POLITICA - Path	COLUMN NAME	PA_L3_OPINION_POLITICA
PA_L3_OPINION_POLITICA - List	KNOWN VALUES	PA_L3_OPINION_POLITICA
PA_L3_AFILIACION_POLITICA - Path	COLUMN NAME	PA_L3_AFILIACION_POLITICA
PA_L3_AFILIACION_POLITICA - List	KNOWN VALUES	PA_L3_AFILIACION_POLITICA
PA_L3_AFILIACION_SINDICAL - Path	COLUMN NAME	PA_L3_AFILIACION_SINDICAL
PA_L3_AFILIACION_SINDICAL - List	KNOWN VALUES	PA_L3_AFILIACION_SINDICAL
PA_FIN_REFERENCIA_CREDITO - Path	COLUMN NAME	PA_FIN_REFERENCIA_CREDITO
PA_FIN_MONTO_DEUDA - Path	COLUMN NAME	PA_FIN_MONTO_DEUDA
PA_FIN_MONTO_DEUDA - Type	DATA TYPE	PA_FIN_MONTO_DEUDA
PA_FIN_INGRESO_MENSUAL - Path	COLUMN NAME	PA_FIN_INGRESO_MENSUAL
PA_FIN_INGRESO_MENSUAL - Type	DATA TYPE	PA_FIN_INGRESO_MENSUAL
PA_FIN_PATRIMONIO - Path	COLUMN NAME	PA_FIN_PATRIMONIO
PA_FIN_PATRIMONIO - Type	DATA TYPE	PA_FIN_PATRIMONIO
PA_FIN_PENSION - Path	COLUMN NAME	PA_FIN_PENSION
PA_FIN_PENSION - Type	DATA TYPE	PA_FIN_PENSION
PA_FIN_SEGURO_SOCIAL - Path	COLUMN NAME	PA_FIN_SEGURO_SOCIAL
PA_FIN_SEGURO_SOCIAL - List	KNOWN VALUES	PA_FIN_SEGURO_SOCIAL
PA_FIN_PROGRAMA_SOCIAL - Path	COLUMN NAME	PA_FIN_PROGRAMA_SOCIAL
PA_FIN_PROGRAMA_SOCIAL - List	KNOWN VALUES	PA_FIN_PROGRAMA_SOCIAL
PA_FIN_NIVEL_SOCIOECONOMICO - Path	COLUMN NAME	PA_FIN_NIVEL_SOCIOECONOMICO
PA_FIN_NIVEL_SOCIOECONOMICO - List	KNOWN VALUES	PA_FIN_NIVEL_SOCIOECONOMICO
PA_FIN_VIVIENDA - Path	COLUMN NAME	PA_FIN_VIVIENDA
PA_FIN_VIVIENDA - List	KNOWN VALUES	PA_FIN_VIVIENDA
PA_PENAL_ANTECEDENTES - Path	COLUMN NAME	PA_PENAL_ANTECEDENTES
PA_PENAL_CAUSA - Path	COLUMN NAME	PA_PENAL_CAUSA
PA_PENAL_CAUSA - Regex	VALUE FORMAT	PA_PENAL_CAUSA